

Wide Area Augmentation of the Global Positioning System

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The Wide Area Augmentation System (WAAS) is being deployed by the Federal Aviation Administration (FAA) to augment the Global Positioning System (GPS). The WAAS will aid GPS with the following three services. First, it will broadcast spread-spectrum ranging signals from communication satellites. The airborne WAAS receiver will add these new ranging signals to the GPS constellation of measurements. By so doing, the augmented position fix will be less sensitive to the failure of individual system components, thus improving time availability and continuity of service. Second, the WAAS will use a nationwide ground network to monitor the health of all satellites over our airspace and flag situations which threaten flight safety. This data will be modulated on to the WAAS ranging signals and broadcast to the users, thereby guaranteeing the integrity of the airborne position fix. Third, the WAAS will use the ground network to develop corrections for the errors which currently limit the accuracy of unaugmented GPS. This data will also be included on the WAAS broadcast and will improve position accuracy from approximately 100 m to 8 m.

When complete, the augmented system will provide an accurate position fix from satellites to an unlimited number of aircraft across the nation. It will be the primary navigation system for aircraft in oceanic routes, enroute over our domestic airspace, in crowded metropolitan airspaces, and on airport approach.

I. INTRODUCTION

The Global Positioning System (GPS) is a satellite-based navigation and time transfer system created by the U.S. Department of Defense (DoD). As a developmental system, GPS has provided approximately ten years of increasing service to military and civilian users in a wide variety of applications. Since 1993, GPS has provided three-dimensional (3-D) positioning worldwide nearly 24 h/day. In September 1993, GPS included 24 satellites and reached Initial Operational Capability (IOC). Today, GPS includes the Standard Positioning Service (SPS), which provides civilian users with 100-m accuracy, and it serves military users with the Precise Positioning Service (PPS) which provides 20-m accuracy.

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In time, GPS will be used for a wide variety of aircraft operations. These will include flight in oceanic routes, en route over our domestic airspace, and in crowded metropolitan airspaces. In addition, GPS will be used on the final approach to airports, which demands the greatest safety and reliability. Airport approach applications will include: nonprecision approach where GPS will be used solely for horizontal positioning, and precision approach where GPS will be used for both horizontal and vertical position fixing. To serve all these applications, GPS must be augmented to meet stringent requirements with respect to accuracy, integrity, continuity, and time availability.

Accuracy specifies the position error at the 95% level in the absence of system failures. The accuracy of stand-alone GPS is approximately 100 m in the horizontal and 150 m in the vertical. This unaugmented accuracy is certainly adequate for aircraft over oceans or far from airports, but is not adequate for precision approach operations, which require 95% accuracies of a few meters or less.

Integrity and continuity are the main measures of flight safety—they address the ability of the navigation system to detect and repair system failures in a timely fashion. Integrity fails when the position error exceeds a certain protection limit *and* this error is not annunciated to the pilot or the aircraft guidance system within a specified time-to-alarm. In contrast, continuity fails when an aircraft operation must be aborted for any unscheduled reason. Continuity failures include obvious failures of the ground or air hardware, which are easily recognized. Clearly, they are not as dangerous as integrity failures, because the pilot or air traffic control has a much greater chance of successful intervention. Nonetheless, continuity requirements are still quite demanding.

For example, if an aircraft is conducting a precision approach, then the probability of a continuity failure must be on the order of 10^{-5} to 10^{-6} for the duration of the approach. The probability of an integrity failure must be on the order of 10^{-7} to 10^{-9} for about the last two minutes of that approach. The time-to-alarm varies from 2 to 6 s

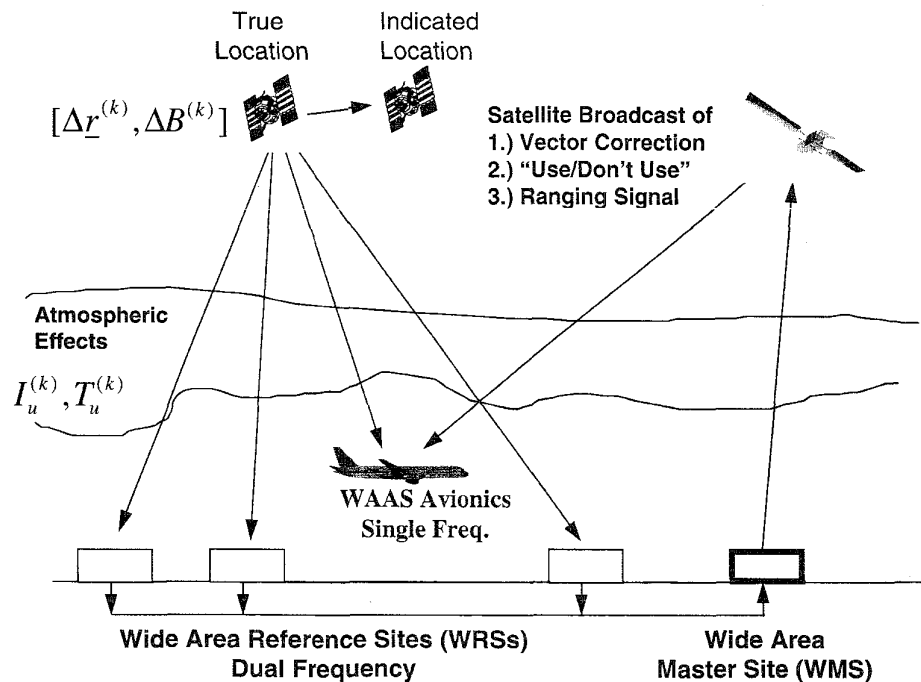


Fig. 1. WAAS. The vector $\Delta \mathbf{r}^{(k)}$ connects the true location of the k th satellite to the satellite location as indicated by the GPS navigation message. The scalar $\Delta B^{(k)}$ is the difference between the true transmission time of the GPS signal and the transmission time indicated by the GPS navigation message. Finally, $I_u^{(k)}$ and $T_u^{(k)}$ are pseudorange measurement errors due to unmodeled ionospheric and tropospheric refraction. In this case, the subscript u denotes the user.

and the protection level varies from approximately 2 to 20 m—all depending on the type of approach.

Finally, *time availability* is the fraction of time for which the system is operational—providing position fixes with the specified accuracy, integrity and continuity. In contrast to integrity and continuity, availability is not intended to measure flight safety. Rather it is a measure of the operational acceptability or economy of a navigation system. For enroute flight down through nonprecision approach, the required availability is 99.999%. For precision approach it is between 99.9% and 99.99%.

As shown in Fig. 1, the Wide Area Augmentation System (WAAS) consists of a ground network that computes and transmits GPS integrity and correction data to users and also provides a ranging signal [1]–[3]. The ground network includes a geographically distributed set of GPS receivers at precisely known reference locations. These receivers continuously monitor all of the GPS satellites, and are known as wide area reference stations (WRS's). An example network of WRS's for the United States is shown in Fig. 2. These WRS's send raw GPS measurements back to redundant wide area master stations (WMS's). One such WMS is also shown in Fig. 1.

The WMS processes the WRS observations to determine four corrections for each monitored satellite. One correction is for the satellite clock, and the other three are for the satellite location. The WMS also estimates a set of corrections for ionospheric delays. If any of these corrections cannot be determined with sufficient confidence,

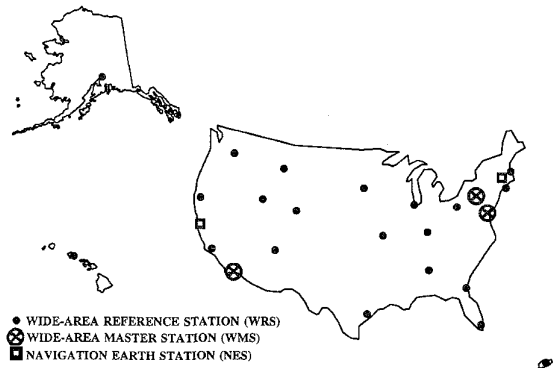


Fig. 2. An example of an WAAS ground network for the United States.

then the WMS does not send the correction. Rather, it sets a *do not use* flag for the troublesome satellite or for the corresponding ionospheric data. This *do not use* flag is an integrity warning. All this correction and integrity data is packed into the WAAS message, which is broadcast to airborne users via the communication satellites.

The WAAS signals carry GPS integrity and correction data to users and also provide ranging signals that augment the primary GPS constellation. Initially, the WAAS signals will be broadcast to users from geostationary communication satellites. In fact, Inmarsat-3 satellites will be the first to carry the WAAS navigation payload, and they are scheduled to be launched in 1996 and 1997. These payloads

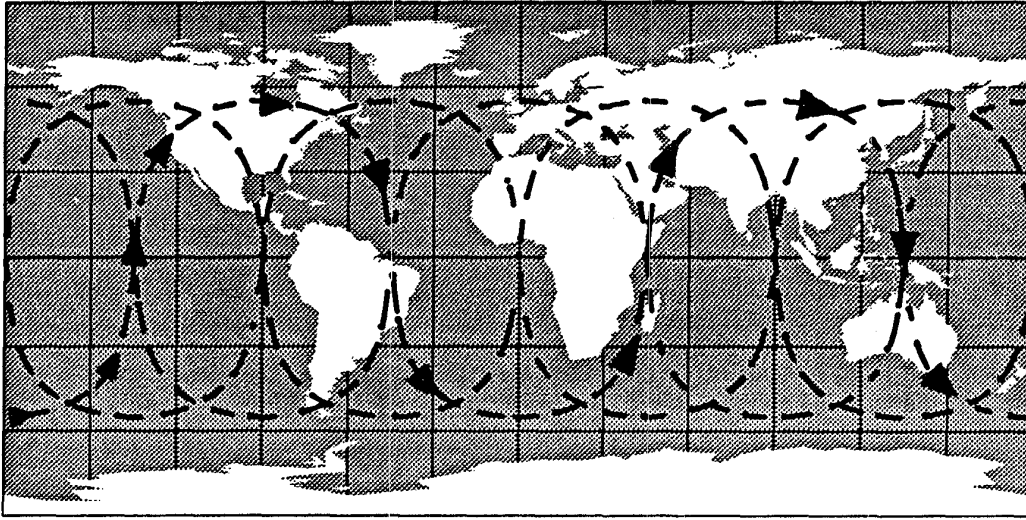


Fig. 3. Ground tracks for four satellites in the GPS constellation. (Reproduced with permission by J. Spilker.)

will broadcast *GPS-like* signals which can be received by slightly modified GPS user equipment. These WAAS ranging signals will augment the ranging signals provided by GPS, and by so doing they will improve the time availability and continuity of the combined service.

The WAAS ranging signals will also carry approximately 250 b/s of data. This capacity will be used to carry the integrity warnings and differential corrections developed by the ground network for all the GPS satellites as well as the WAAS communication satellites. It will allow the WAAS to warn all aircraft in the coverage area of integrity threats within 6 s. The WAAS messages will also send the differential corrections often enough to provide a vertical positioning accuracy of better than 8 m.

In summary, WAAS will augment the GPS so that it can be used as the primary navigation sensor for aircraft in oceanic routes, enroute over our domestic airspace, and in crowded metropolitan airspaces. In addition, it will be approved for use on nonprecision approach and precision approach.

Sections II and III of this paper review the fundamental operation of GPS and the WAAS, respectively. Specifically, Section III describes the ranging function of the WAAS signal and quantizes the improvement in position fixing availability made possible by the WAAS ranging sources. It also discusses nonprecision approach, where the WAAS data need only provide coarse corrections and integrity. Then it discusses precision approach, where the WAAS must provide high-precision vector corrections to achieve the accuracy requirements. Finally, Section III provides accuracy and availability predictions for the WAAS.

Sections IV–VI detail the WAAS ground processing, and are based on an experimental wide area network developed at Stanford University for the FAA. Section IV describes the WRS's. Section V describes the development of the ionospheric corrections which are included in the WAAS

broadcast to the user. Section VI describes the generation of the GPS satellite clock and location corrections.

Section VII summarizes the WAAS data format. Section VIII describes the WAAS/GPS avionics, and includes a discussion of the weighted navigation solution and autonomous fault detection. A set of WAAS flight trials are described in Section IX, and Section X contains a brief summary.

This paper describes the current state of the art with respect to wide area augmentation of the GPS. However, it represents only the starting point for the many exciting wide area developments yet to come.

II. FUNDAMENTALS OF GPS

Stand-alone GPS uses 24 satellites placed asymmetrically in six orbital planes, where each plane is inclined by 55° relative to the equatorial plane. The satellites are approximately 26 560 km above the center of the earth, with an orbital period of 12 h and repeating ground tracks as shown in Fig. 3. The orbital period is measured in sidereal-time, where a sidereal day is the period of earth's rotation relative to the fixed stars (not the sun) and is approximately four minutes shorter than a solar day. Hence, the noontime subsatellite point slowly drifts in its fixed ground track from day to day.

A. GPS Signals

Each satellite continuously broadcasts direct-sequence, spread-spectrum signals on two frequencies: $f_{L1} = 1575.42$ MHz and $f_{L2} = 1227.60$ MHz. The GPS signal design is only summarized here because an authoritative description can be found in [4]. The signal received from each GPS satellite is

$$s_{L1}(t) = \sqrt{2C_X}D(t)X(t)\sin(2\pi f_{L1}t + \theta_{L1}) + \sqrt{2C_P}D(t)P(t)\cos(2\pi f_{L1}t + \theta_{L1}) \quad (1)$$

$$s_{L2}(t) = \sqrt{2C_{L2}}D(t)P(t)\cos(2\pi f_{L2}t + \theta_{L2}). \quad (2)$$

As shown, the L_1 signal has an in-phase component and a quadrature component, where both are modulated by navigation data ($D(t)$). These two components are received with power C_X and C_P , respectively. The L_2 signal has only one component, which is received with power C_{L2} . The L_2 component is also modulated with the navigation data. This navigation data is developed by the GPS ground control segment and predicts the satellite's location and clock offset (relative to GPS system time and the universal coordinated time (UTC)). The errors in these location and clock predictions are small. Even so, they must be corrected by the WAAS in real-time to achieve the accuracy required for precision approach. The data rate for the navigation information ($D(t)$) is only 50 b/s.

Each GPS satellite also modulates its transmission with a pair of spread-spectrum codes ($X(t)$ and $P(t)$). The code $X(t)$ modulates one of the L_1 components and is known as the clear acquisition (C/A) code. It modulates the amplitude of the carrier with a pseudorange sequence of ± 1 at a modulation rate of 1.023 MHz. In spread-spectrum parlance, this modulation rate is known as the chipping rate and each chip is $1/1.023 \mu\text{s}$ in duration. As such, the C/A code spreads the bandwidth of the L_1 component by a factor of 20 000 relative to the bandwidth required for the data stream alone. Another code ($P(t)$) modulates the other L_1 component and the L_2 signal. This code also multiplies the carrier by a sequence of ± 1 , but at a higher 10.23 MHz chipping rate. In this case, the bandwidth expansion is 200 000. In contrast to the C/A code, $P(t)$ can be switched to a secure code, which is only known to users authorized by the U.S. DoD.

Spread spectrum is vital to the function of GPS [4], [5]. It enables precise ranging measurements in the presence of noise, reflected signals and interfering signals. The C/A code is the basis of the *Standard Positioning Service* (SPS), which is for civilian use. The faster code is the basis of the *Precise Positioning Service* (PPS), which is solely for authorized users. Since the WAAS is for international aviation use, the airborne WAAS receiver use the C/A code only. However, the WAAS ground segment does include receivers which use both frequencies.

Spread spectrum also allows all of the GPS satellites to transmit continuously on the same two frequencies. The correlation properties of the codes allow the receiver to distinguish between the 5–12 GPS signals which are typically received at any instant of time [6]. The C/A code for each satellite is a unique member of a family of length 1023 Gold sequences [4]. The P code for each satellite is a unique segment of a very long pseudorandom sequence. These codes are chosen such that a GPS receiver can track all the satellites in view without confusion. The WAAS leverages this property of the GPS codes, and uses other sequences from the same family of Gold codes for its transmissions. As described later, the WAAS broadcasts continuously on f_{L1} with a signal very similar to the L_1 component of GPS.

B. GPS Measurements and Accuracy

For each of several satellites, the user equipment measures a *pseudorange* and demodulates the navigation message. Let (x_u, y_u, z_u) and $(x^{(k)}, y^{(k)}, z^{(k)})$ denote the user's location and the k th satellite's location in 3-D space. The measured arrival time of the spread-spectrum code is denoted $\tau_u^{(k)}$, where the superscript k denotes the k th satellite and the subscript u denotes the user measurement. This measurement is given by

$$\begin{aligned} \tau_u^{(k)} &= r_u^{(k)} + (b_u - B^{(k)}) + \nu_u^{(k)} \\ &= \sqrt{(x_u - x^{(k)})^2 + (y_u - y^{(k)})^2 + (z_u - z^{(k)})^2} \\ &\quad + (b_u - B^{(k)}) + \nu_u^{(k)}. \end{aligned} \quad (3)$$

The measured pseudorange is equal to the true range ($r_u^{(k)}$) plus an unknown bias equal to the difference between the receiver clock and GPS system time ($b_u - B^{(k)}$).

GPS pseudorange measurements also suffer from a variety of measurement errors which are combined in $\nu_u^{(k)}$. Modern GPS receivers can achieve a range measurement *precision* of around 0.5 m using the C/A code. However, the largest error in C/A code measurements is a slowly varying bias intentionally introduced by the DoD for reasons of national security. This error is called *selective availability* (SA) and introduces a ranging error of approximately 30 m. SA has a decorrelation time of approximately 180 s.

The next largest SPS errors are introduced by ionospheric and tropospheric refraction. Even though SPS receivers model these atmospheric effects, the residual errors for the ionosphere and troposphere can easily be 10 and 2 m, respectively. Since SA is significantly larger than the typical atmospheric effects, the total variance of the measurement error is approximately 30 m.

The GPS receiver uses pseudorange measurements to four or more well placed satellites to solve for the user's 3-D position (x_u, y_u, z_u) and clock offset ($b_u - B^k$). A GPS position solution is based on a linearized version of (3). Let $\delta x_u, \delta y_u, \delta z_u, \delta b_u$ be departures from the best *a priori* estimate of x_u, y_u, z_u, b_u , and let $\delta \rho^{(k)}$ be the corresponding perturbation in the pseudorange. In addition, let the position coordinates be the ordered right-hand set: East, North, and Up. Further, let the satellite azimuth ($az^{(k)}$) be measured as clockwise from true North and elevation ($el^{(k)}$) be measured up from the local horizontal. If the position and time errors are small, then linearized equations for K satellites can be written as (see (4) at the top of the following page).

If $\mathbf{x}_u$ denotes the perturbation $[\delta x_u, \delta y_u, \delta z_u, \delta b_u]^T$ and $\boldsymbol{\rho} = [\delta \rho^{(1)}, \delta \rho^{(2)} \dots \delta \rho^{(K)}]^T$, then we can simply write

$$G\mathbf{x}_u = \boldsymbol{\rho} \quad (5)$$

where G is determined solely by user-satellite geometry and is called the *geometry* matrix.

If $K \geq 4$, then these *navigation equations* can usually be solved to provide an optimal least squares estimate of the user's position and clock offset. If the covariance matrix describing the pseudorange measurement errors is P_ρ , then

$$\begin{bmatrix} \cos(el^{(1)}) \sin(az^{(1)}) & \cos(el^{(1)}) \cos(az^{(1)}) & \sin(el^{(1)}) & 1 \\ \cos(el^{(2)}) \sin(az^{(2)}) & \cos(el^{(2)}) \cos(az^{(2)}) & \sin(el^{(2)}) & 1 \\ \vdots & \vdots & \vdots & \vdots \\ \cos(el^{(K)}) \sin(az^{(K)}) & \cos(el^{(K)}) \cos(az^{(K)}) & \sin(el^{(K)}) & 1 \end{bmatrix} \begin{bmatrix} \delta x_u \\ \delta y_u \\ \delta z_u \\ \delta b_u \end{bmatrix} = \begin{bmatrix} \delta \rho^{(1)} \\ \delta \rho^{(2)} \\ \vdots \\ \delta \rho^{(K)} \end{bmatrix} \quad (4)$$

the maximum likelihood estimate of the perturbation in the user position and clock is

$$\hat{\mathbf{x}}_u = A_U \boldsymbol{\rho} \quad (6)$$

where

$$A_U = [G^T P_\rho^{-1} G]^{-1} G^T P_\rho^{-1}$$

and the corresponding error in the estimate of $\mathbf{x}_u$ has covariance matrix given by

$$P_x = [G^T P_\rho^{-1} G]^{-1}. \quad (7)$$

If the GPS measurements are not corrected, then the individual pseudorange errors are dominated by SA. All of the pseudorange errors have approximately equal standard deviation $\sigma_\rho^{(k)} \approx \sigma_\rho$, where σ_ρ^2 is the common variance of the pseudorange measurement errors. In this case, $P_\rho \approx \sigma_\rho^2 I$ where I is the identity matrix, and the maximum likelihood estimate is given by

$$\hat{\mathbf{x}}_u = [G^T G]^{-1} G^T \boldsymbol{\rho} \quad (8)$$

and the error in the estimate has the following covariance:

$$P_x = [G^T G]^{-1} \sigma_\rho^2. \quad (9)$$

As shown in (9), the position error suffered by a GPS user is proportional to the pseudorange measurement error (σ_ρ^2), but it also depends on the geometry of the user and the satellites through the matrix $[G^T G]^{-1}$. If the satellites are clustered from the user's point of view, the geometry is poor, and poor accuracies result. An ideal geometry would be three low-elevation satellites 120° apart in azimuth, with a fourth satellite at zenith. The measures which describe the goodness of the geometry are known as dilution-of-precision (DOP) values. These are now defined.

When the position coordinates are the ordered right hand set (East, North, and Vertical), then

$$P_x = \begin{bmatrix} E_{\text{DOP}}^2 & & & \\ & N_{\text{DOP}}^2 & & \\ & & V_{\text{DOP}}^2 & \\ & & & T_{\text{DOP}}^2 \end{bmatrix} \sigma_\rho^2.$$

As shown, East DOP (E_{DOP}) is the element in the upper left of the $[G^T G]^{-1}$ and so forth for North DOP, vertical DOP, and time DOP. An estimate of any position or time error can now be obtained by multiplying the appropriate DOP value by the variance of the pseudorange measurement error.

The most commonly used measures of *accuracy* for vertical, horizontal, and 3-D positioning are

$$\begin{aligned} 2\sigma_V &= 2V_{\text{DOP}}\sigma_\rho \\ 2\sqrt{\sigma_E^2 + \sigma_N^2} &= 2\sqrt{(N_{\text{DOP}})^2 + (E_{\text{DOP}})^2}\sigma_\rho \\ &= 2H_{\text{DOP}}\sigma_\rho \\ 2\sqrt{\sigma_E^2 + \sigma_N^2 + \sigma_V^2} &= 2\sqrt{(N_{\text{DOP}})^2 + (E_{\text{DOP}})^2 + (V_{\text{DOP}})^2} \\ &\quad \cdot \sigma_\rho \\ &= 2P_{\text{DOP}}\sigma_\rho \end{aligned}$$

where these equations also define horizontal DOP (or H_{DOP}) and positional DOP (or P_{DOP}). The specified accuracies are achieved approximately 95% of the time [7].

The geometric dilution-of-precision values defined above depend on user location and on time, where the latter dependence arises because GPS satellites are moving relative to the surface of the earth. The GPS constellation has been designed to give low dilution-of-precision when averaged over all locations worldwide and over the day. The worldwide median daylong values of P_{DOP} and H_{DOP} are 2.0 and 1.2, respectively. As mentioned earlier, the standard deviation of the pseudorange measurement errors are approximately 30 m when the C/A code is used. At any time, if P_{DOP} and H_{DOP} are equal to their median values, then the 3-D and horizontal accuracies are 120 m and 70 m, respectively. (The specified 95% horizontal accuracy of the Standard Positioning Service is 100 m.)

III. FUNDAMENTALS OF WAAS

The WAAS improves accuracy in two fundamental ways.

- 1) It reduces range measurement error by sending *differential corrections* for each GPS satellite, thus decreasing σ_ρ^2 . The WAAS correction is capable of reducing the pseudorange measurement error from around 30 m to approximately 1 or 2 m (1σ).
- 2) It improves geometry by adding new ranging signals to the set of available GPS measurements, thus increasing the probability that the diagonal elements of $[G^T G]^{-1}$ are small.

A. WAAS Signals

Each WAAS signal will be given by

$$s(t) = \sqrt{2C} X(t) D(t) \cos(2\pi f_{L1} t + \theta). \quad (10)$$

The WAAS signals will use the GPS f_{L1} carrier frequency and will be modulated with data ($D(t)$) which carries

integrity warnings and differential corrections for each GPS satellite. In addition, each signal is also modulated with a spread-spectrum code ($X(t)$). Like the GPS C/A code, this code is a ± 1 pseudorange sequence with a chipping rate of 1.023 MHz.

The spread-spectrum code serves the same functions for WAAS as it does for GPS. It enables precise ranging measurements to be made even in the presence of noise, reflected signals and interfering signals. It also allows the WAAS to share the GPS f_{L1} frequency with the GPS signals. In fact, the WAAS codes have been chosen from the same family of codes as the GPS C/A codes. GPS only reserved 32 codes from this family of 1025 possibilities, and so many codes remained for WAAS. Hence, the receiver will be able to distinguish the WAAS signals from the GPS signals using the same spread-spectrum mechanism that is used to separate the GPS signals from each other.

B. Impact of WAAS Ranging

The WAAS ranging capability significantly improves the percentage of time that accurate position fixes are available. This ranging function is possible, because the phase of the WAAS code is synchronized to GPS time [8]. The resulting improvement is shown in Fig. 4, which shows the percent of time that a given level of vertical accuracy ($\sigma_v = V_{DOP}\sigma_p$) is available. This time availability calculation averages over a sidereal day since the GPS satellite positions repeat every 24 h. It also averages over all possible user locations in the conterminous United States (CONUS).

First, the *Stand-Alone GPS* curve shows the availability of vertical accuracy when only the GPS constellation is considered in the computation of $(G^T G)^{-1}$. It assumes that all pseudorange errors have $\sigma_p = 32$ m. As shown, the vertical accuracy is 150 m ($1\sigma_v$) or better, approximately 99.9% of the time over CONUS.

Second, the *Stand-Alone GPS and Inmarsat* curve shows the availability of a given level of vertical accuracy when the planned Inmarsat-3 satellites are added to the constellation and broadcast WAAS ranging signals. This curve assumes that the Inmarsat-3 satellites are operating at their planned locations of the Atlantic Ocean Region West (55° W), Atlantic Ocean Region East (15° W), and Pacific Ocean Region (180°). Inmarsat-3 will also have a satellite over the Indian Ocean, but this satellite is not visible in CONUS and does not influence the computations. With the addition of these new ranging sources, a vertical accuracy of 150 m or better is now available 99.99% of the time.

The *Stand-Alone GPS and Inmarsat* curve assumes that the GPS pseudoranges are uncorrected and still have $\sigma_p = 32$ m. In contrast, the Inmarsat ranging capability will not be degraded by SA; hence $\sigma_p \approx 2$ m for the Inmarsat satellites. As discussed below, the variance of pseudorange errors for signals which are not degraded or which have been corrected by WAAS is a function of elevation angle, and a weighted least squares estimate is used.

Both the *Stand-Alone GPS* and the *Stand-Alone GPS and Inmarsat* curves assume that the GPS pseudoranges are uncorrected. The two curves on Fig. 4 marked *Inmarsat*

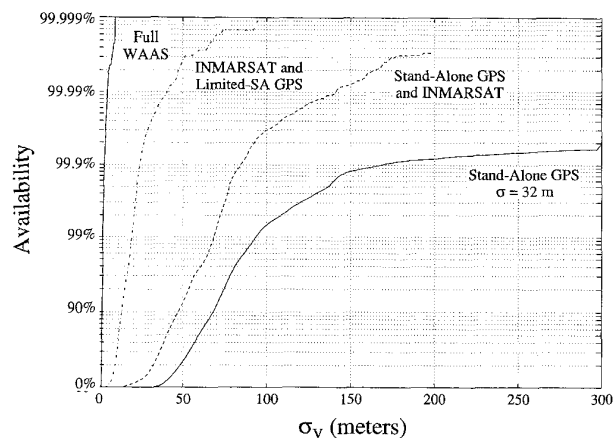


Fig. 4. Availability of vertical position accuracy over CONUS. With the exception of the *Stand-Alone GPS* curve, these curves assume that the planned Inmarsat-3 satellites are used to broadcast WAAS ranging signals.

and *Limited-SA GPS* and *Full WAAS* assume differential corrections of increasing accuracy. These will be discussed in the following subsections, which describe nonprecision approach and precision approach, respectively. Fig. 4 incorporates the effect of possible satellite failures, where GPS satellite failures are based on the model described in [9] and WAAS satellite failures are based on the model described in [10].

C. Nonprecision Approach and Coarse Corrections

When an aircraft is conducting a nonprecision approach to an airport, the pilot typically uses a baro-altimeter for vertical position information and a ground-based radio navigation aid for horizontal information. The horizontal information is typically derived from any of the following terrestrial sources: a VHF omnidirectional radio (VOR) navigation aid, distance measuring equipment (DME), Loran-C, automatic direction finding (ADF) to a nondirectional beacon (NDB), or a "localizer-only" transmission from an instrument landing system (ILS).

GPS receivers can also be certified for nonprecision approach if they conform to the performance requirements in the FAA's Technical Standard Order C-129, which was released in 1992 [11]. These receivers still derive vertical information from a baro-altimeter, but GPS now provides the horizontal information. These TSO-C129 receivers are based on GPS alone, and do not require the use of WAAS. However, they are for supplemental use only, which means that another source of horizontal guidance, such as VOR/DME, must be available on the aircraft. Such a backup is necessary, because the fraction of time that GPS alone provides adequate accuracy and integrity is too small. If adequate GPS service is not available, then the pilot can revert to the other positioning system.

The time availability of horizontal positioning accuracy using GPS plus WAAS is shown in Fig. 5. Today the horizontal accuracy ($2\sqrt{\sigma_E^2 + \sigma_N^2}$) required for nonprecision approach is only 560 m. However, the aviation community

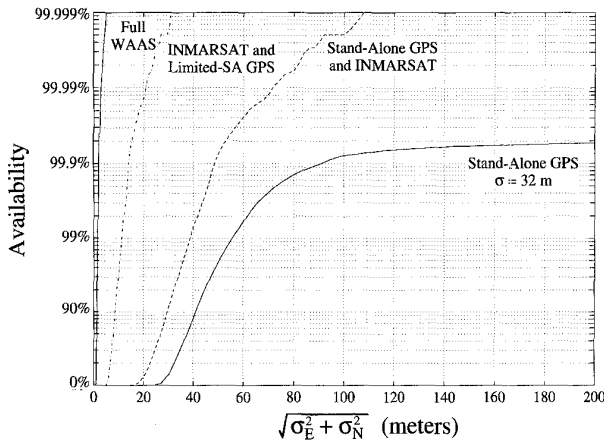


Fig. 5. Availability of horizontal position accuracy over CONUS. With the exception of *Stand-Alone GPS*, the curves assume the broadcast of WAAS ranging signals from the planned Inmarsat-3 satellites.

desires a horizontal accuracy of 100 m to be available 99.999% of the time. As shown, the ranging capability of the Inmarsat-3 satellites combined with uncorrected GPS is marginal for this latter objective. However, if the Inmarsat ranging signals also carry coarse corrections for GPS to reduce the effects of SA, then the availability goal can be met. In fact, the *Inmarsat and Limited-SA GPS* curve assumes that the coarse corrections improve the pseudorange measurement error for GPS from $\sigma_p = 32$ to $\sigma_p = 16$ m. Once again, the weighted least squares navigation solution described later is used, because the WAAS ranging signals from the Inmarsat satellites have smaller measurement errors than the GPS satellites (even with coarse corrections).

A dense network of reference stations is not required to generate the coarse corrections for nonprecision approach because the corrections need only be moderately accurate. In fact, 10–12 reference stations may well provide adequate worldwide coverage [12]. However, denser networks, like the one shown in Fig. 2, are required to provide the accuracy required for precision approach.

D. Precision Approach and Vector Corrections

Unlike nonprecision approach, precision approach is based on a constant rate of descent resulting in a smooth *glide path*. This glide path typically makes an angle of 3° with the horizontal.

Precision approach is divided into three categories that correspond to different weather conditions. Specifically, the three categories are distinguished from each other by the *decision height* (DH) and the *runway visual range* (RVR). The decision height is the altitude at which the pilot must decide whether or not to complete the landing. Runway visual range quantifies visibility along the runway. Category I conditions exist when the decision height is at least 200 ft and the RVR is at least 2400 ft. Category II conditions exist when the decision height is between 100 and 200 ft and the RVR is at least 1200 ft. Category III conditions

exist when the decision height is less than 100 ft or the RVR is less than 1200 ft.

Today, the ILS provides all three categories of service. However, ILS is aging, and the FAA must replace these installations with equivalent or better service over the next ten years. Hopefully, an equally safe system with substantially lower cost for operation and maintenance can be substituted. In addition, the FAA would like to provide a rudimentary precision approach capability to the thousands of airports in this country which currently have no precision approach instrumentation at all.

Approximately 110 airports nationwide require either Category II or III landing guidance. These landing systems must provide vertical accuracies of 1.2 and 0.6 m ($2\sigma_v$) for Categories II and III, respectively. In the future, some variety of local area DGPS system (with a reference station close to the runway) will probably be used to guide the aircraft, because the Category II and III vertical accuracy requirements are beyond the capability of any WAAS. Suitable differential GPS systems are being developed by the FAA as part of their Local Area Augmentation System (LAAS) Program. A particularly imaginative system based on the GPS carrier phase and *pseudo-satellites* has been developed at Stanford University [13].

In contrast, over 1000 U.S. runways currently have Category I ILS. If the decision height is 200 ft, then the vertical accuracy requirement is between 4.1 and 7.6 m ($2\sigma_v$) 99.9% of the time [15]. These Category I accuracy requirements can be met using either *scalar* corrections over a local area or *vector* corrections over larger areas.

A local area differential GPS system is depicted in Fig. 6, and such systems can certainly provide Category I capability [15]. As mentioned in the Introduction, wide area systems send multiple corrections for each satellite and separate corrections for the ionosphere. In contrast, a local area system simply sends a scalar correction for each satellite. This scalar is valid provided that the aircraft is close to the reference station. A scalar correction will necessarily degrade as the aircraft moves away from the reference site, since a single number cannot represent the nominal error as well as its spatial gradient. The error sources which cause the correction to decorrelate spatially are: ionospheric refraction, tropospheric refraction, and errors in the GPS ephemeris information included in the GPS navigation message. This spatial decorrelation is detailed in [16].

While local area DGPS could be used to provide a nationwide Category I capability, it would require a ground installation at most of the approximately 1000 Category I airports in the U.S. Local siting is required to insure the validity of the scalar correction, and because the broadcast system is usually a line-of-sight radio. Such an extensive infrastructure would be expensive to procure, operate and maintain. A system which could provide Category I service to a large area (hundreds of miles across) without installations at every airport would be much more cost effective. Such a wide area system must use multiple corrections for each satellite to capture the spatial variability of the error.

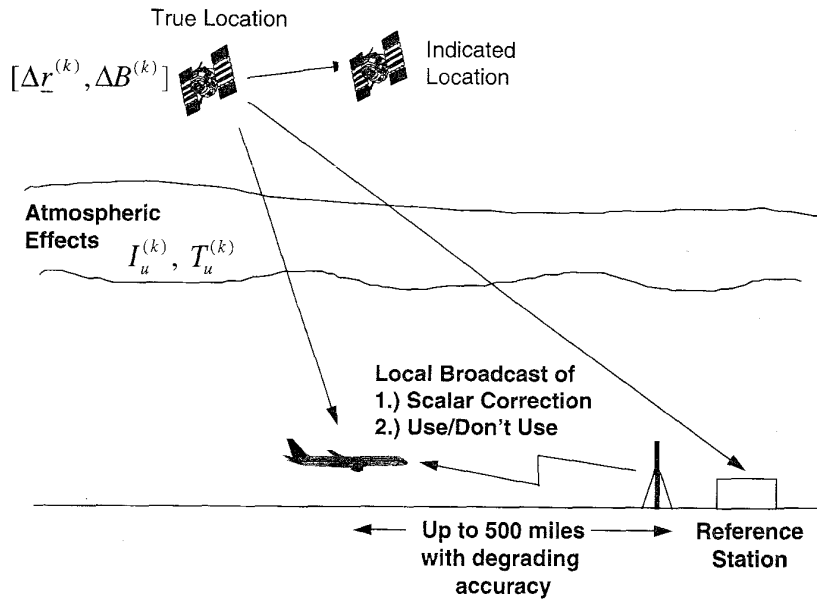


Fig. 6. Local Area Broadcast of Scalar Corrections. In contrast to wide area systems, local area differential GPS broadcasts a single correction for each satellite.

Vector corrections contain the following components for each satellite:

1) *Satellite Clock Offset*: This correction estimates the error in the transmission time of the GPS signal and does not decorrelate spatially. In other words, the satellite clock offset is independent of the user location. However, a clock correction does decorrelate temporally due to SA. In fact, the satellite clock corrections must be sent every 6–20 s, which is much more frequent than any other component of the vector correction.

2) *Satellite Location*: This correction estimates the error in the satellite location as derived from the GPS navigation message. Estimates of all three location components ($\Delta \mathbf{r}_k$) are sent, where this vector connects the true location of the k th satellite to the location given by the satellite's navigation message. In component form, this data does not decorrelate spatially and decorrelates very slowly in time. In fact, the update rate for the location data is not set by concerns over accuracy. Rather, it is set to insure that the time to first position fix for a new user entering the WAAS is not too large.

The vector corrections also carry separate data which describe the ionosphere. This data may be a grid of ionospheric delay estimates or it may be a set of coefficients for an orthonormal function description of the ionospheric delay. This data does not decorrelate rapidly in time, and an update interval of 2–5 min is probably adequate. Once again, the update rate for this correction is set to minimize the time-to-first-fix for a new user.

E. Predictions of WAAS Accuracy and Availability

This subsection takes two approaches to predict the accuracy and availability of the WAAS with full precision vector corrections, where the two approaches are

summarized in Fig. 7. Both approaches recognize that full-precision WAAS corrections greatly reduce errors due to SA. Consequently, the pseudorange errors after correction depend on the elevation angle of the satellite and no longer have equal variance. Correspondingly, both analyses assume that the user equipment employs a weighted least-squares approach to estimate position. Specifically, the covariance of the error in the user's position and clock estimates are given by

$$P_x = [G^T P_\rho^{-1} G]^{-1}$$

$$\sigma_v = \sqrt{[(G^T P_\rho^{-1} G)^{-1}]_{3,3}}$$

where the $[A]_{3,3}$ denotes the element in the third row and third column of any matrix A . As before, the matrix P_ρ is the covariance of the pseudorange measurement errors after correction.

Two classes of components for the pseudorange errors exist. The first includes errors which are generated in the neighborhood of a GPS receiver and thus cannot be ameliorated using the WAAS. The second class includes all errors in the WAAS corrections themselves. More specifically, the covariance of the pseudorange error after correction is assumed to be

$$P_\rho = P_{\rho, \text{local}} + P_{\rho, \text{WAAS}}.$$

The local errors and the errors in the WAAS correction are now discussed in turn.

1) *Local Errors*: Local errors are shown in Fig. 7 and include receiver noise and interference, receiver multipath, and tropospheric model residuals. These errors appear at the WAAS reference stations as well as the mobile user receiver. In both cases, the errors are assumed to be zero

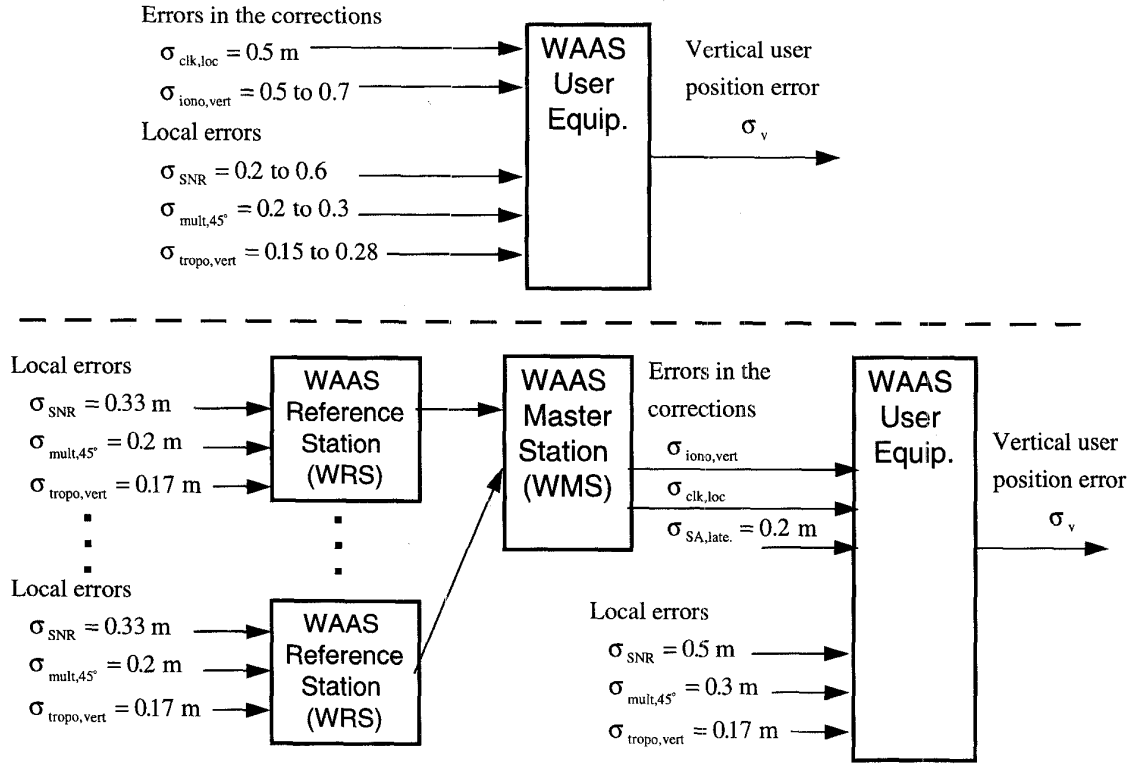


Fig. 7. Two approaches for the analysis of WAAS accuracy. The top approach assumes that the variance of the errors in the WAAS corrections can be adequately approximated based on our flight results. The bottom approach estimates the variance of the correction errors by propagating WRS measurement errors through the reference and master station processing.

mean and uncorrelated, hence

$$P_{\rho, \text{local}} = \begin{bmatrix} (\sigma_{\rho, \text{local}}^{(1)})^2 & 0 & 0 & 0 \\ 0 & (\sigma_{\rho, \text{local}}^{(2)})^2 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & (\sigma_{\rho, \text{local}}^{(K)})^2 \end{bmatrix}$$

$$(\sigma_{\rho, \text{local}}^{(k)})^2 = \sigma_{\text{SNR}}^2 + \frac{\sigma_{\text{multi,45}}^2}{\tan^2(e\ell)} + \frac{\sigma_{\text{tropo,vert}}^2}{\sin^2(e\ell)}. \quad (11)$$

In (11), (σ_{SNR}) is due to natural noise and multiple access interference at the user receiver. Typical values vary between 0.2 and 0.6 m depending on the user equipment. The next term $(\sigma_{\text{multi,45}}^2 / \tan^2(e\ell))$ is the standard deviation of the pseudorange error due to multipath at the user receiver. As shown, it is a function of elevation angle, and typical values for $\sigma_{\text{multi,45}}$ lie between 0.2 and 0.3 m. Finally, $(\sigma_{\text{tropo,vert}}^2 / \sin^2(e\ell))$ is due to the tropospheric refraction which the user is unable to model. Typical values for $\sigma_{\text{tropo,vert}}$ vary between 0.15 and 0.28. This error is also a function of elevation angle, and tropospheric obliquity is approximately $1/\sin(e\ell)$.

2) *Errors in the WAAS Corrections:* Errors in the WAAS corrections are also shown in Fig. 7 and include residual ionospheric errors and residual clock/location errors. Our two approaches differ with respect to the estimation of these errors.

Our first approach is shown at the top of Fig. 7. It simply assumes that the errors in the correction components are uncorrelated and have variances which can be approximated or bounded based on WAAS field trials. Specifically, the covariance of the errors in the WAAS correction is assumed to be

$$P_{\rho, \text{WAAS}} = \begin{bmatrix} (\sigma_{\rho, \text{WAAS}}^{(1)})^2 & 0 & 0 & 0 \\ 0 & (\sigma_{\rho, \text{WAAS}}^{(2)})^2 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & (\sigma_{\rho, \text{WAAS}}^{(K)})^2 \end{bmatrix}$$

$$(\sigma_{\rho, \text{WAAS}}^{(k)})^2 = \sigma_{\text{clk,loc}}^2 + F(e\ell)\sigma_{\text{iono,vert}}^2. \quad (12)$$

The term $(\sigma_{\text{clk/loc}}^2)$ is the variance of the error in the satellite clock and location corrections. This standard deviation is typically 0.5 m. The next term $F(e\ell)\sigma_{\text{iono,vert}}^2$ is the variance of the error in the ionospheric delay after correction. It is equal to the variance of the vertical ionospheric delay $(\sigma_{\text{iono,vert}}^2)$ times the obliquity factor or thickness of the ionosphere $(F(e\ell))$ given that the satellite is at an elevation angle of $e\ell^\circ$. The standard deviation of the vertical ionospheric error $(\sigma_{\text{iono,vert}})$ is typically between 0.5 and 0.7 m. In addition, ionospheric obliquity varies between one and three.

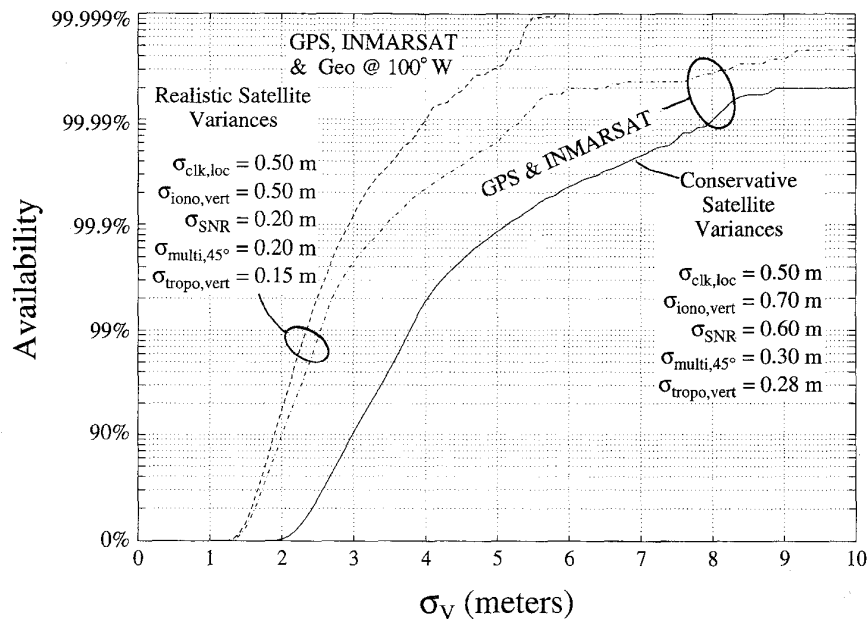


Fig. 8. Availability of vertical position accuracy over CONUS. This analysis is for ranging WAAS satellites located at the planned Inmarsat-3 locations.

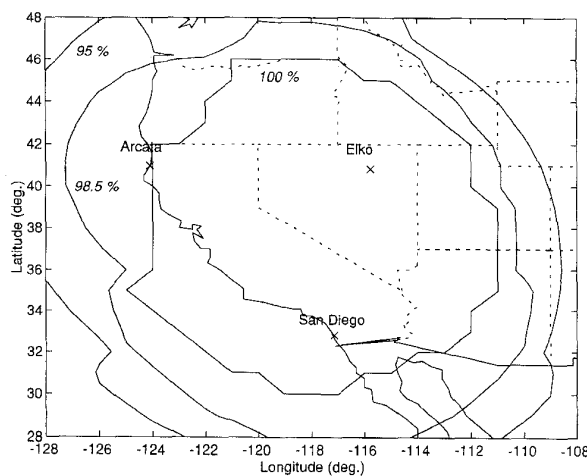


Fig. 9. Predicted time availability of a vertical accuracy of 7.6 m.

These assumptions have been used to generate the performance curves marked *Full WAAS* in Figs. 4 and 5 and all the curves in Fig. 8. As shown in Fig. 8, if the conservative values of the variances listed above are used, then a vertical error of 7.6 m ($2\sigma_v$) is available 99% of the time. If values which seem realistic based on the flight trials described later are used, then the same accuracy is available more than 99.9% of the time. As shown, adding a geostationary satellite at the middle of the North American arc (100°) improves the availability to 99.99%.

In contrast to the simple approach described above, our second approach propagates the local measurement errors at the reference stations through the reference and master station processing. These errors (noise, multipath, and troposphere) propagate through the WAAS and cause

errors in the ionospheric corrections ($\sigma_{\text{iono,vert}}$) and the satellite clock and location corrections ($\sigma_{\text{clk,loc}}$). These correction errors are added to the errors, which are local to the user receiver, to yield a total error for each corrected pseudorange measurement. In turn, these propagate through the user's navigation solution and manifest themselves as errors in the user position estimate.

This error propagation requires linear models for the reference and master station processing, and these models will be described in Sections IV–VIII of this paper as well as [25]. It also requires assumptions for the covariance of the ionospheric errors, the satellite clock errors and the satellite location errors in the absence of WAAS corrections. A full description of these assumptions is beyond the scope of this introductory paper, but they are well described in [25].

In Fig. 7, the local errors suffered at the WAAS reference stations have smaller standard deviations than the local errors at the user receiver. This difference reflects our experience with the survey quality GPS receivers to be used at the WRS's. The aircraft receiver standard deviations are in accord with the currently drafted requirements for certifiable WAAS avionics.

Additionally, the aircraft receiver is assumed to suffer from an additional error ($\sigma_{SA,late}$) due to the latency in the clock correction data. This error is assumed to have zero mean and to be uncorrelated from pseudorange to pseudorange. It is due to the random acceleration of SA, and the standard deviation is assumed to be 0.20 m. This deviation would be typical for WAAS, where the latency is around 10 s.

Fig. 9 shows the time availability of a vertical accuracy of 7.6 m ($2\sigma_v$) as a function of user location. It is for the sparse

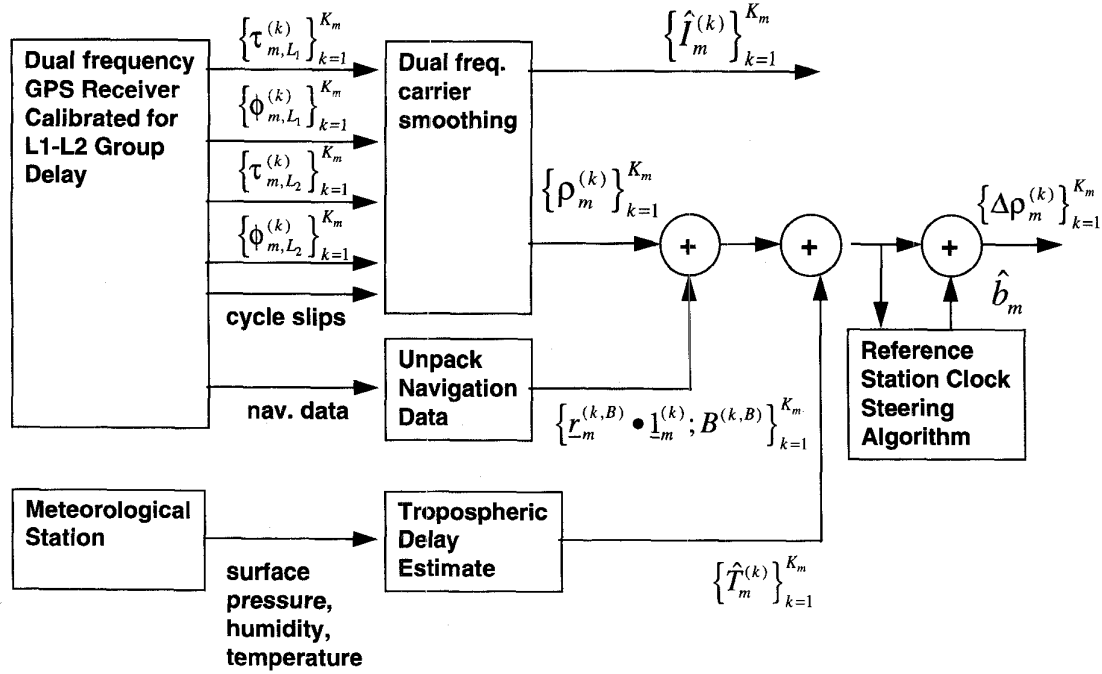


Fig. 10. Wide area reference station (WRS).

wide area network established by Stanford University for the FAA with WRS's in Elko, NV, and San Diego and Arcata, CA. As shown, a vertical accuracy of 7.6 m is available 100% of the time for a region roughly defined by the reference stations, where this assessment is based on 6000 samples. The availability of 7.6 m accuracy begins to decrease fairly rapidly outside of the region spanned by the reference stations.

IV. WIDE AREA REFERENCE STATIONS (WRSS)

As discussed in the Introduction, a wide area differential GPS network includes wide area reference stations (WRS's), which are distributed throughout the area where service is to be provided. A single WRS uses a dual frequency receiver to produce code and carrier phase observations at the GPS frequencies f_{L1} and f_{L2} as shown in Fig. 10. The set of code observations at f_{L1} and f_{L2} are denoted $\{\tau_{m,L1}^{(k)}\}_{k=1}^{K_m}$ and $\{\tau_{m,L2}^{(k)}\}_{k=1}^{K_m}$, respectively, where K_m is the number of satellites in view of the m th reference station. Similarly, the carrier observations are denoted $\{\phi_{m,L1}^{(k)}\}_{k=1}^{K_m}$ and $\{\phi_{m,L2}^{(k)}\}_{k=1}^{K_m}$, respectively. As before, the superscript k denotes the satellite and the subscript m denotes the m th reference station.

The measured code and carrier observations at the two frequencies are well modeled by

$$\tau_{m,L1}^{(k)} = (r_m^{(k)} + b_m - B^{(k)}) + I_{m,L1}^{(k)} + T_m^{(k)} + \nu_{m,\tau}^{(k)} \quad (13)$$

$$\phi_{m,L1}^{(k)} = (r_m^{(k)} + b_m - B^{(k)}) - I_{m,L1}^{(k)} + T_m^{(k)} + N_{m,L1}^{(k)} \lambda_{L1} + \nu_{m,\phi}^{(k)} \quad (14)$$

$$\tau_{m,L2}^{(k)} = (r_m^{(k)} + b_m - B^{(k)}) + I_{m,L2}^{(k)} + T_m^{(k)} + \nu_{m,\tau}^{(k)} \quad (15)$$

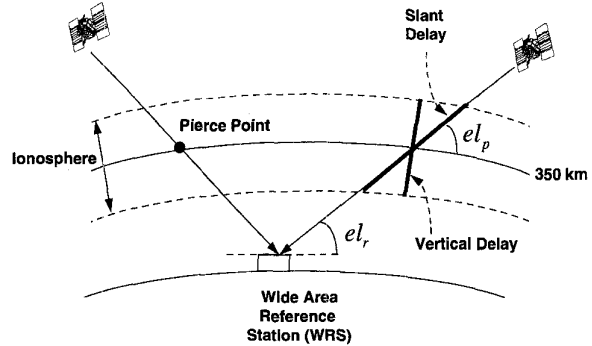


Fig. 11. Ionospheric observation geometry.

$$\phi_{m,L2}^{(k)} = (r_m^{(k)} + b_m - B^{(k)}) - I_{m,L2}^{(k)} + T_m^{(k)} + N_{m,L2}^{(k)} \lambda_{L2} + \nu_{m,\phi}^{(k)} \quad (16)$$

The code and carrier phase observations share the same pseudorange $(r_m^{(k)} + b_m - B^{(k)})$. All four measurements suffer from noise (ν), which is assumed to be zero mean and uncorrelated. The carrier phase has much less noise than the code phase, but the carrier phase measurements are ambiguous. After all, the phase lock loops output measurements which are fractions of a carrier cycle, while the number of whole cycles contained in the measurement is unknown. This ambiguity is represented by $N_m^{(k)} \lambda$ in the equations above.

As shown in (13)–(16), all of the measurements from the k th satellite include tropospheric delay ($T_m^{(k)}$). The troposphere is not dispersive, and hence this delay is the same for all four measurements.

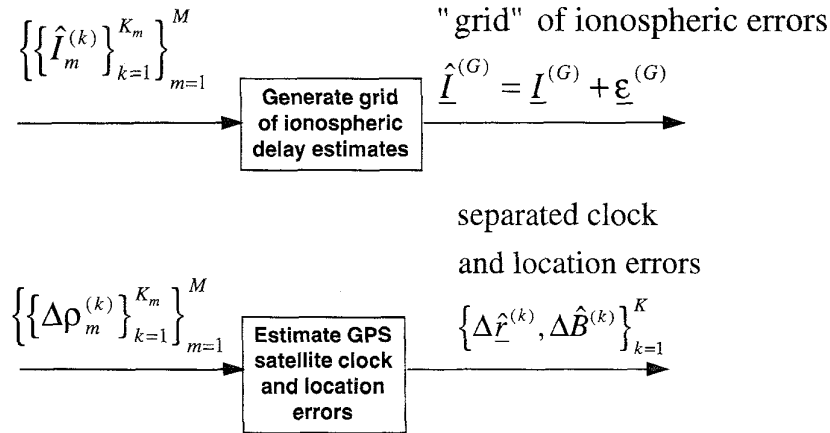


Fig. 12. WMS algorithms.

In contrast, the ionosphere is dispersive. In fact, ionospheric delay is approximately

$$I_m^{(k)} = 40.3 \text{ TEC} / c f^2$$

where TEC is the total electron content of the column which surrounds the ray from the satellite to the WRS, c is the speed of light, and f is the frequency of the signal. Since the ionospheric delay is different for the two GPS transmission frequencies, measurements at f_{L1} and f_{L2} can be used to estimate the delay with high accuracy [17].

The ionosphere also causes the carrier and code to diverge. Specifically, the code phase is determined by the group velocity (v_g) and the carrier phase is determined by the phase velocity (v_p). These two quantities are related by $c = \sqrt{v_g v_p}$, where c is the speed of light. Consequently, ionospheric refraction will cause the carrier phase to advance and the code phase to retard. These phase changes are equal but opposite; and this effect is known as code-carrier divergence.

A simple filter uses the *changes* in the carrier measurements to estimate changes in the pseudorange and ionosphere [18]. Carrier phase differences are used since they do not suffer from the $N_m^{(k)} \lambda$ ambiguity. Moreover, the carrier phase difference is used to smooth the estimates of the pseudorange and ionosphere—with the goal of reducing the effects of noise, multipath and interference. This dual-frequency carrier-smoothing filter is based on the following observation equation.

$$\begin{bmatrix} \tau_{m,L1}^{(k)} \\ \delta\phi_{m,L1}^{(k)} \\ \tau_{m,L2}^{(k)} \\ \delta\phi_{m,L2}^{(k)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{40.3}{cf_{L1}^2} & 0 \\ 0 & 1 & 0 & \frac{-40.3}{cf_{L1}^2} \\ 1 & 0 & \frac{40.3}{cf_{L2}^2} & 0 \\ 0 & 1 & 0 & \frac{-40.3}{cf_{L2}^2} \end{bmatrix} \begin{bmatrix} \rho_m^{(k)} \\ \delta\rho_m^{(k)} \\ \text{TEC} \\ \delta\text{TEC} \end{bmatrix} + \nu_m^{(k)}$$

where $\rho_m^{(k)} \approx r_m^{(k)} + b_m - B^{(k)} + T_m^{(k)}$ is the smoothed pseudorange estimate. Note that it is the differencing of carrier measurements in time ($\delta\phi_m^{(k)}$) which provide very accurate estimates of the change in the pseudorange as well

as the change in the ionosphere—yet do not suffer from the $N_m^{(k)} \lambda$ ambiguity.

This filter provides estimates of the delay introduced along the slanted path through the ionosphere. The WRS's convert the slant ionospheric delays to vertical delay estimates by dividing by the ionospheric obliquity:

$$F(el) = \frac{1}{\sin(el_p)} = \frac{1}{\sqrt{1 - \left(\frac{r_e \cos(el_r)}{r_e + h_m} \right)^2}} \quad (17)$$

where el is the elevation angle for the k th satellite. As shown in Fig. 11, el_p is the local angle at the corresponding pierce point, el_r is the elevation angle at the WRS, r_e is the average radius of the earth, and h_m is the height of the maximum electron density (assumed to be 350 km). These vertical ionospheric delay estimates are denoted $\{\hat{I}_m^{(k)} = I_m^{(k)} + \epsilon_m^{(k)}(el_m)\}_{k=1}^{K_m}$, where $\hat{I}_m^{(k)}$ and $I_m^{(k)}$ are the estimated and true vertical delays. The errors $\epsilon_m^{(k)}(el_m)$ depend on the elevation angles of the satellites as seen by the WRS. These estimates are sent to the WMS for further processing as described in Section V.

The smoothed pseudorange estimates are well-modeled by

$$\begin{aligned} \rho_m^k &= (r_m^{(k)} + b_m - B^{(k)}) + \Delta I_m^{(k)} + T_m^{(k)} + \nu_m^{(k)} \\ &\approx (r_m^{(k)} + b_m - B^{(k)}) + T_m^{(k)} + \nu_m^{(k)}. \end{aligned}$$

Through an abuse of notation, the second line assumes that the residual ionospheric error is small enough to be included in the noise term ($\nu_m^{(k)}$). This equation is essentially the same as our equation for the code phase measurement τ_m^k . However, the ionospheric component has been removed, and ν_m^k is generally much smaller than $\nu_{m,r}^k$ because carrier smoothing has attenuated the effects of receiver noise, interference and multipath.

The reference station subtracts the nominal range to the satellite (as computed from the data in the satellite navigation message and the known location of the reference station) from the measured pseudorange. It also subtracts the nominal satellite clock offset contained in the navigation

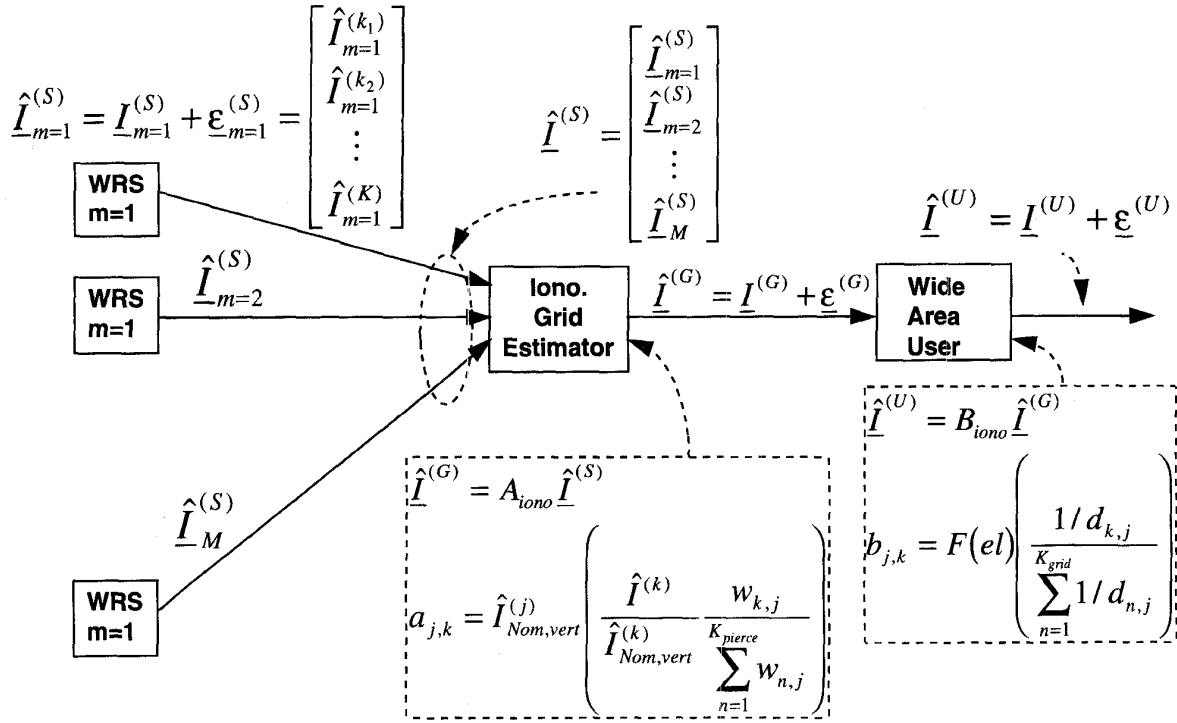


Fig. 13. Overview of ionospheric processing.

message. These reduced pseudoranges are given by

$$\begin{aligned} \Delta \rho_m^{(k)} &= r_m^{(k)} - r_m^{(k,B)} + b_m - (B^{(k)} - B^{(k,B)}) + T_m^{(k)} \\ &\quad + \nu_m^{(k)} \\ &\approx \Delta \mathbf{r}^{(k)} \bullet \mathbf{1}_m^{(k)} + b_m - \Delta B^{(k)} + T_m^{(k)} + \nu_m^{(k)} \end{aligned}$$

where $\Delta \mathbf{r}^{(k)}$ is the vector which connects the true location of the satellite to the location of the satellite according to the navigation message. In the above, $\mathbf{1}_m^{(k)}$ denotes the unit vector from the k th satellite toward the m th reference station. Additionally, $B^{(k)}$ is the true offset of the satellite transmission from GPS time, and $B^{(k,B)}$ is the offset according to the navigation message. Consequently, $\Delta B^{(k,B)}$ is the satellite clock error and contains the effects of SA.

Next, the reference station subtracts an estimate of the tropospheric delay from the observed pseudorange. Importantly, the tropospheric estimate is not based on elevation angle alone. Rather, it is based on elevation angle and local measurements of pressure, temperature and humidity. Indeed, a model without this side information can suffer errors of 2 m for satellites at low elevation angles. In contrast, a model with surface measurements does not usually have errors greater than 30 cm [19].

Finally, the reference station of Fig. 10 uses a *clock steering* algorithm to estimate the reference station clock offset. The sole goal of this algorithm is to control the size of the pseudorange residuals before they are transmitted to the WMS. It does this by adding a constant to all of the measurements where this constant is a rough approximation of the reference station clock offset from GPS time. As

discussed in Section VI, the WMS uses common view time transfer to provide a much more accurate estimate of the differences in the reference station clocks.

The local reference station clock estimates are subtracted from the smoothed pseudoranges to yield

$$\Delta \rho_m^{(k)} = \Delta \mathbf{r}^{(k)} \bullet \mathbf{1}_m^{(k)} + \Delta b_m - \Delta B^{(k)} + \nu_m^{(k)}. \quad (18)$$

As shown in Fig. 10, these pseudorange residuals (for all the satellites in view) are sent along with the vertical ionospheric delay estimates $\{\hat{I}_m^{(k)}\}_{k=1}^{K_m}$ to the WMS for processing.

The WMS houses two main estimators as shown in Fig. 12. The first, described in the next section, uses the ionospheric delay data from the WRS's to generate ionospheric delay estimates for a *grid* of locations which uniformly cover the service area. The second is described in Section VI, and uses the *iono* and *tropo-free* pseudorange residuals to estimate the clock and location errors for each satellite in view of the network.

V. REAL-TIME ESTIMATION OF IONOSPHERIC DELAYS

An overall block diagram of the ionospheric data processing is shown in Fig. 13. As described in the previous section, the wide area reference stations include dual frequency GPS receivers to measure the slant ionospheric delays to all satellites in view. These delays are measured over a widely dispersed set of *pierce points*. For example, the pierce points at one instant of time for Stanford's experimental network are shown in Fig. 14.

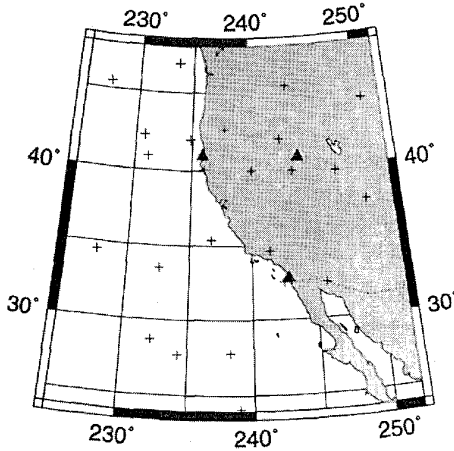


Fig. 14. Ionospheric pierce points at one instant of time for Stanford's experimental wide area experimental network.

The WMS receives ionospheric data from all of the reference stations, and this data is denoted

$$\hat{\mathbf{I}}^{(S)} = \begin{bmatrix} \hat{I}^{(k=1)} \\ \hat{I}^{(k=2)} \\ \vdots \\ \hat{I}^{(k=K_{\text{pierce}})} \end{bmatrix} = \mathbf{I}^{(S)} + \boldsymbol{\epsilon}^{(S)}$$

where the index $\{k = 1, 2, \dots, K_{\text{pierce}}\}$ now counts all of the pierce points for all of the satellites as seen by all of the reference stations. The true ionospheric delay at the pierce points is denoted $\mathbf{I}^{(S)}$ and the errors in the reported values are denoted $\boldsymbol{\epsilon}^{(S)}$.

In general, this data could be processed to provide any number of ionospheric delay models for broadcast to the users. For example, it could be used to find the coefficients of a two-dimensional (2-D) polynomial description of ionospheric delay. In this paper, we follow the lead of [20]–[22] and use the reference station data to form a uniform *grid* of ionospheric estimates. The grid has points every 5 or 10° in latitude and longitude and is broadcast to the users. The user receivers interpolate between the grid points to find estimates of the ionospheric delay at their pierce points.

We assume that the grid point estimates are given by

$$\hat{\mathbf{I}}^{(G)} = \mathbf{A}_{\text{iono}} \hat{\mathbf{I}}^{(S)}.$$

In other words, the estimates are a linear function of the incoming measurements. In our WAAS testbed, the elements of $\mathbf{A}_{\text{iono}}$ have been chosen in the following heuristic fashion [21]:

$$a_{j,k} = \hat{I}_{\text{nom},v}^{(j)} \left(\frac{\hat{I}^{(k)}}{\hat{I}_{\text{nom}}^{(k)}} \frac{w_{k,j}}{\sum_{n=1}^K w_{n,j}} \right)$$

where $a_{j,k}$ is the element in the j th row of the k th column of $\mathbf{A}$, which corresponds to the j th grid point and the k th pierce point. These elements use a nominal model for the vertical ionospheric delays at the j th grid point ($\hat{I}_{\text{nom},v}^{(j)}$)

and the k th pierce point ($\hat{I}_{\text{nom},v}^{(k)}$). They also use a weight $w_{k,j}$ for the contribution of the k th pierce point to the j th grid point.

The nominal model ($\hat{I}_{\text{nom},v}^{(j)}$, $\hat{I}_{\text{nom},v}^{(k)}$) is used to capture the expected variation of the ionosphere with time and geomagnetic latitude and longitude. To date, the well-known Klobuchar single frequency model has been used [23]. The Klobuchar model is also the standard used by single frequency stand-alone GPS receivers. It estimates the ionospheric delay as a function of geomagnetic latitude, time of day, and some parameters which are contained in the navigation message from the GPS satellite. In uncorrected applications, it typically reduces the ionospheric error by a factor of two.

The measurement weights ($w_{k,j}$) generally include two effects. First, they decrease as the expected correlation of the ionospheric delay at the k th pierce point with the delay at the j th grid point decreases. Second, the weights decrease as the quality of the ionospheric measurement at the k th pierce point decreases. If two pierce points are equally close to a grid point, then greater weight should generally be given to the pierce point corresponding to a satellite at higher elevation angle. Weights which blend these considerations are currently being investigated.

The simple assignment $w_{k,j} = 1/d_{k,j}$ has been used to generate Fig. 15, where $d_{k,j}$ is simply the distance from the k th pierce point to the j th grid point. Fig. 15 shows the total vertical ionospheric delay over the Western United States for one instant of time, and was generated by the Stanford experimental network shown on the underlying map.

VI. REAL-TIME ESTIMATION OF THE GPS CLOCK AND LOCATION ERRORS

As shown in Fig. 12, the WMS receives the following *iono* and *tropo-free* pseudorange residuals from the reference stations

$$\{\{\Delta\rho_m^{(k)} = \Delta\mathbf{r}^{(k)} \bullet \mathbf{1}_m^{(k)} + \Delta b_m - \Delta B^{(k)} + \nu_m^{(k)}\}_{k=1}^{K_m}\}_{m=1}^M.$$

In this equation, the $\Delta\mathbf{r}^{(k)}$ and the $\Delta B^{(k)}$ are the estimanda of interest. Estimates of these parameters will be broadcast to the airborne users. In contrast, the user has no interest in the reference station clock offsets (the Δb_m). However, these *nuisance* parameters must still be estimated by the central station to yield an accurate estimate of the data of interest. Finally, $\nu_m^{(k)}$ is the measurement noise.

In general, the GPS clock and location can be estimated by using a real-time version of the well-known and refined orbit estimators described in [26]. These orbit determination techniques have the important advantage of incorporating accurate models for the satellite motion. After one or two satellite passes, these algorithms provide accurate estimates of the Keplerian orbit parameters (or *ephemeris*) of the satellite. In addition, they do not use the ephemerides in the GPS navigation message, and as such are not vulnerable to errors or transients in the GPS message.

For the time being, the Stanford WMS uses the simpler strategy, shown in Fig. 16, which estimates the satellite *location* errors directly as follows:

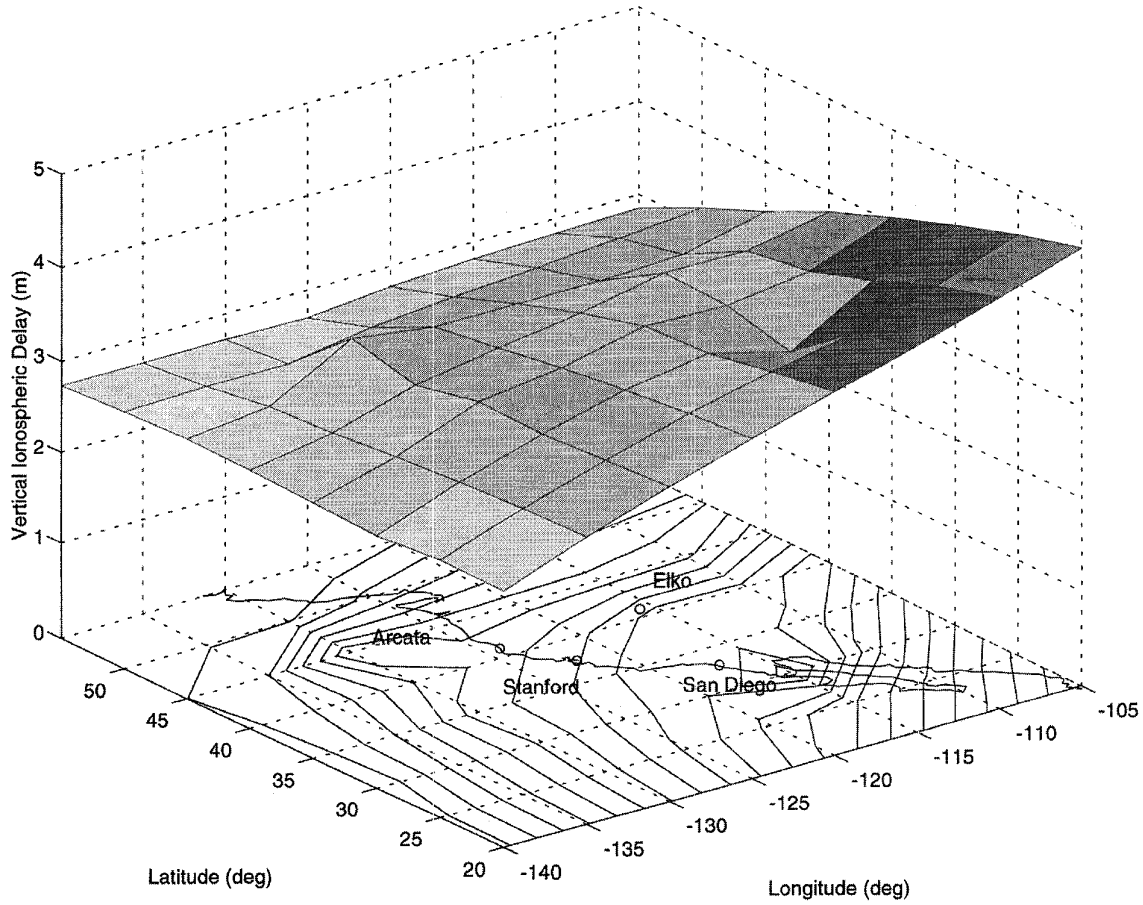


Fig. 15. Ionospheric grid for one instant of time as generated by Stanford's experimental wide area network.

- Common view time transfer is used to remove the WRS clock offsets from the measurements. After the WRS clock offsets are removed, the observation equations are uncoupled. In other words, the observations which depend on the clock and location errors of the k th satellite do not depend on the clock or location errors of any other satellite.
- A single difference is then used to remove the satellite clock offsets from the measurements. At this point, the reduced observations only depend on the location errors ($\Delta \mathbf{r}^{(k)}$) for the given satellite.
- A simple linear estimator is used to estimate the three components of satellite location error. If the satellite of interest is only in view of three or fewer WRS's, then a minimum norm solution is used. If the satellite is in view of four or more WRS's, then an *a priori* covariance for the satellite location error is included in the estimator's cost function.
- The location error estimates are subtracted from the original observations, which are then used to estimate the satellite clock offsets.

Interestingly, this approach is a *snapshot* estimator which takes no advantage of well-understood models of GPS satellite dynamics. As such, it does not provide an accurate estimate of the satellite ephemeris. However, it does provide

an estimate of the satellite location which is valid for users located inside the reference stations. Current research [26] strives to improve the overall performance of WAAS by incorporating satellite dynamical information.

A. Common View Time Transfer

The WMS uses common view time transfer to estimate the difference between the reference station clocks and the master clock located at the central station. Common view time transfer simply differences the pseudorange residuals from two sites as follows:

$$\begin{aligned}\Delta \rho_{m_1}^{(k_1)} &= \Delta \mathbf{r}^{(k_1)} \bullet \mathbf{1}_{m_1}^{(k_1)} + \Delta b_{m_1} - \Delta B^{(k_1)} \\ &\quad + \nu_{m_1}^{(k_1)} \\ \Delta \rho_M^{(k_1)} &= \Delta \mathbf{r}^{(k_1)} \bullet \mathbf{1}_M^{(k_1)} + \Delta b_M - \Delta B^{(k_1)} \\ &\quad + \nu_M^{(k_1)} \\ \Delta \rho_{m_1}^{(k_1)} - \Delta \rho_M^{(k_1)} &= \Delta \mathbf{r}^{(k_1)} \bullet (\mathbf{1}_{m_1}^{(k_1)} - \mathbf{1}_M^{(k_1)}) + \Delta b_{m_1} \\ &\quad - \Delta b_M + \nu_{m_1, M}^{(k_1)}.\end{aligned}$$

As shown, the reference station clock difference appears along with measurement noise and the inner product of the satellite location error and the line-of-sight difference vectors.

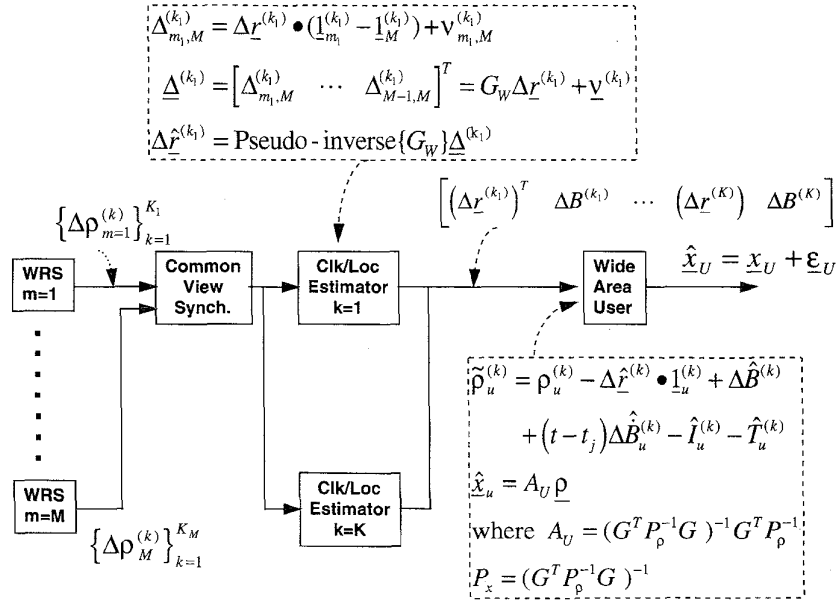


Fig. 16. Overview of clock and location processing.

If the line of sight difference is small and/or the satellite location error is small, then an accurate estimate of the clock difference can be achieved as follows:

$$\begin{aligned} \Delta \hat{b}_{m_1, M} &= \Delta \hat{b}_{m_1} - \Delta \hat{b}_M \\ &= \Delta \rho_{m_1}^{(k_1)} - \Delta \rho_M^{(k_1)} - [\Delta r^{(k_1)} \bullet (\mathbf{1}_{m_1}^{(k_1)} - \mathbf{1}_M^{(k_1)}) \\ &\quad + \nu_{m_1, M}^{(k_1)}] \\ &\approx \frac{1}{K_{m_1, M}} \sum_{k=1}^{K_{m_1, M}} \Delta \rho_{m_1}^{(k_1)} - \Delta \rho_M^{(k_1)} \\ &\approx \frac{1}{K_{m_1, M}} \sum_{k=1}^{K_{m_1, M}} h(t_j) \\ &\quad * [\Delta \rho_{m_1}^{(k_1)}(t_j) - \Delta \rho_M^{(k_1)}(t_j)] \end{aligned}$$

where $K_{m_1, M}$ is the number of satellites in common view of reference stations m_1 and M . Our estimator averages the clock difference as computed from all of the satellites in the common view of the two stations. As such, it reduces the error introduced by the measurement noise and the location errors. In addition, our estimator averages over time to take advantage of the well-behaved drifts of the rubidium clocks at our WRS's. We speculate that this algorithm estimates the clock offsets to within 10 or 20 cm (a fraction of a nanosecond) after the initial transients have died out.

This family of estimated clock offsets $\{\Delta \hat{b}_{m, M}\}_{m=1}^{M-1}$ is very important. It is used as follows to eliminate the clock differences in the observations from the reference stations:

$$\begin{aligned} \Delta \tilde{\rho}_{m_1}^{(k_1)} &= \Delta \rho_{m_1}^{(k_1)} - \Delta \hat{b}_{m_1, M} \\ &\approx \Delta r^{(k_1)} \bullet \mathbf{1}_{m_1}^{(k_1)} + \Delta B_M^{(k_1)} + \nu_{m_1}^{(k_1)} \end{aligned}$$

where $\Delta B_M^{(k_1)}$ is the clock error for satellite k_1 relative to the WAAS master clock. These synchronized pseudorange residuals are the fundamental inputs to the SV location and SV clock estimation algorithms described in the next

subsections. Importantly, the residuals for the different satellites are now decoupled. Consequently, the WMS contains one small estimator for every satellite in view of the network, where each of these estimators is only solving for four unknowns. This situation is shown in Fig. 16. In contrast, if the residuals are not synchronized, then all of the observations are coupled, and a large estimator must solve for tens of unknowns.

The remainder of this section describes two clock/location estimators. This first is used when the satellite of interest is in view of three or fewer reference stations. The second is used when four or more reference stations observe the satellite.

B. Three or Fewer Reference Stations

If the satellite is seen by only one reference station (WRS), then the WMS simply sets the satellite position correction to zero ($\Delta r^{(k_1)} = 0$) and sets the satellite clock correction equal to the synchronized pseudorange residuals. As such, a satellite observed by one WRS only receives a scalar correction. However, this satellite is only in view of users in the same area as the reference station, so the correction will tend to be valid. As described in Section III-B, it will also be given low weight in the user's navigation solution because the satellite has a low elevation angle.

If the satellite is seen by two or three WRS's, then the WMS begins by differencing pairs of synchronized pseudorange residuals as follows:

$$\begin{aligned} \Delta \tilde{\rho}_{m_1}^{(k_1)} &= \Delta r^{(k_1)} \bullet \mathbf{1}_{m_1}^{(k_1)} - \Delta B_M^{(k_1)} + \nu_{m_1}^{(k_1)} \\ -\Delta \tilde{\rho}_M^{(k_1)} &= -(\Delta r^{(k_1)} \bullet \mathbf{1}_M^{(k_1)} - \Delta B_M^{(k_1)} + \nu_M^{(k_1)}) \\ \Delta r_{m_1, M}^{(k_1)} &= \Delta r^{(k_1)} \bullet (\mathbf{1}_{m_1}^{(k_1)} - \mathbf{1}_M^{(k_1)}) + \nu_{m_1, M}^{(k_1)} \end{aligned}$$

This set of differences is no longer a function of the satellite clock offsets; so the 3-D location error for each

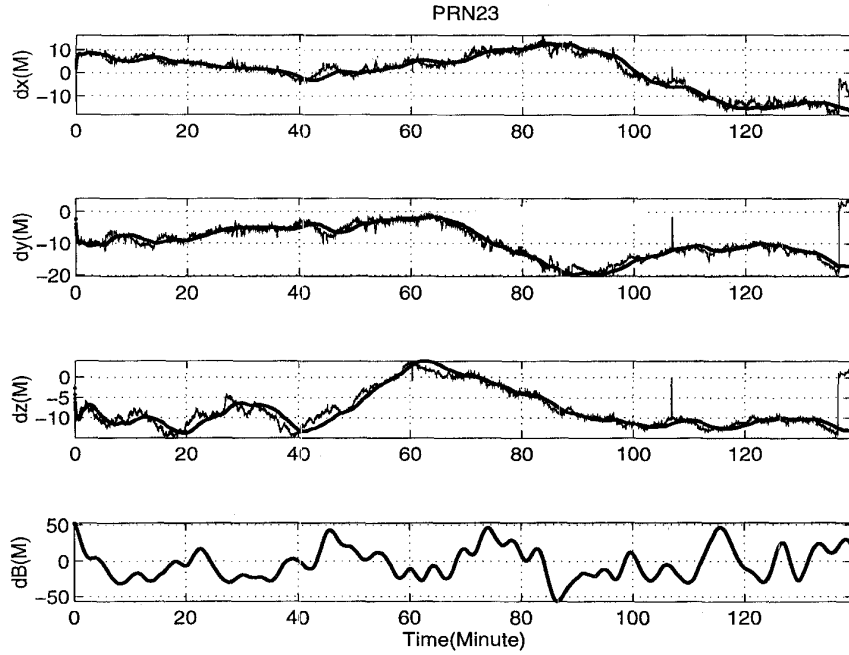


Fig. 17. Corrections for the location and clock of GPS Satellite 23. The top three traces (dx , dy , and dz) are corrections for the spatial coordinates contained in $\Delta\hat{\mathbf{r}}^{(k)}$ and are measured in meters. The bottom trace (dB) is the clock correction ($\Delta B^{(k)}$) and is also measured in meters. In all cases, the k th satellite was GPS satellite 23, which was in view of all three Stanford reference station for the 120 min shown in the plots. The noisier traces are from the snapshot, minimum-norm algorithm described in Section VI-B. The smoother traces are from a filter which uses a simple kinematic model to smooth the corrections.

satellite may be estimated directly using the following set of observation equations:

$$\begin{bmatrix} \Delta_{m_1, M}^{(k_1)} \\ \vdots \\ \Delta_{M-1, M}^{(k_1)} \end{bmatrix} = \begin{bmatrix} (\mathbf{1}_{m_1}^{(k_1)} - \mathbf{1}_M^{(k_1)})^T \\ \vdots \\ (\mathbf{1}_{M-1}^{(k_1)} - \mathbf{1}_M^{(k_1)})^T \end{bmatrix} \Delta\mathbf{r}^{(k_1)} + \boldsymbol{\nu}^{(k_1)}. \quad (19)$$

This linear system of equations can be written more compactly as

$$\Delta\hat{\mathbf{r}}^{(k_1)} = G_W \Delta\mathbf{r}^{(k_1)} + \boldsymbol{\nu}^{(k_1)}.$$

Since the satellite is observed by three or fewer reference stations ($M < 4$), the system is underspecified. In other words, more than one state ($\Delta\mathbf{r}^{(k_1)}$) will satisfy the system of equations. From these, we choose the solution that has minimum length (or energy) among the satisfactory solutions. If $(G_W G_W^T)^{-1}$ exists, then the minimum norm solution is given by

$$\Delta\hat{\mathbf{r}}_{\text{MN}}^{(k_1)} = G_W^T (G_W G_W^T)^{-1} \Delta^{(k_1)}$$

where the subscript MN denotes minimum norm. Among the set of solutions which yield zero measurement residuals, this solution minimizes

$$J_{\text{MN}} = [\Delta\mathbf{r}^{(k_1)}]^T \Delta\mathbf{r}^{(k_1)}.$$

If the *a priori* covariance of the satellite position error is $\Lambda_r^{(k_1)} = E\{\Delta\mathbf{r}^{(k_1)}[\Delta\mathbf{r}^{(k_1)}]^T\}$, a weighted cost function may be used

$$J_{\text{WMN}} = [\Delta\mathbf{r}^{(k_1)}]^T [\Lambda_r^{(k_1)}]^{-1} \Delta\mathbf{r}^{(k_1)}$$

where the subscript WMN denotes weighted minimum norm. The corresponding estimator is given by

$$\begin{aligned} \Delta\hat{\mathbf{r}}_{\text{WMN}}^{(k_1)} &= \Lambda_r^{(k_1)} G_W^T (G_W [\Lambda_r^{(k_1)}]^{-1} G_W^T)^{-1} \Delta^{(k_1)} \\ &= A_W \Delta^{(k_1)}. \end{aligned}$$

This weighted minimum norm solution is important to wide area DGPS because the $\Delta\mathbf{r}^{(k_1)}$ vector can be rotated to along-track, cross track, and radial coordinates, where these have unequal *a priori* variances. Specifically, the along-track error tends to be the largest.

After the satellite position is estimated, this estimate is used in the synchronized pseudorange residual to obtain an estimate of the satellite clock error. In other words, any portion of the synchronized residual which is not attributed to satellite location error is assumed to be due to SA.

The estimator described in this subsection has been used to generate the clock and location error and error rate estimates shown in Figs. 17 and 18. In Fig. 17, the clock and location error estimates for the GPS satellite designated PRN 23 are shown. The noisier traces are directly from the snapshot, minimum-norm algorithm described above. The smoother traces use a filter with a simple kinematic model to filter the data [27]. In Fig. 18, the clock and location error rate estimates are shown. In this case, only the outputs of the smoother are shown. Importantly, the rates of the location error estimates are much smaller than the rates of the clock error estimate. Consequently, the location error estimates can be sent much less frequently than the clock

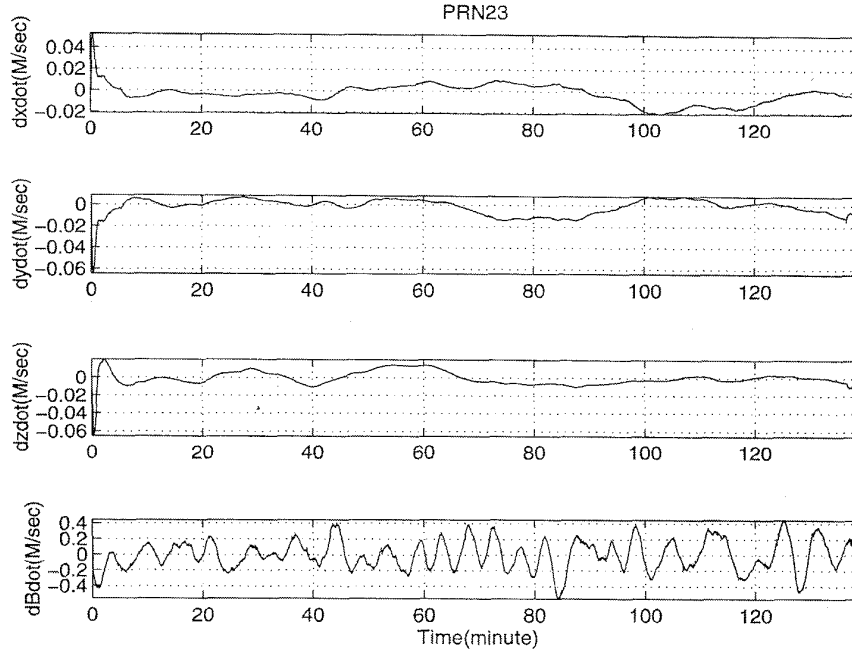


Fig. 18. Correction rates for the location and clock of GPS Satellite 23. The top three traces ($\dot{x}$, $\dot{y}$, and $\dot{z}$) are the rate of change of the corrections for the spatial coordinates contained in $\Delta \mathbf{r}^{(k)}$. These are measured in meters per second. The bottom trace ($\dot{B}$) is the rate of change for the clock correction ($\Delta B^{(k)}$) also measured in meters per second. In all cases, the k th satellite was GPS satellite 23, which was in view of all three Stanford reference station for the 120 min shown in the plots. The noisier traces are from the snapshot, minimum-norm algorithm described in Section VI-B. The smoother traces are from a filter which uses a simple kinematic model to smooth the corrections.

error estimates—recall this is a fundamental goal of wide area processing.

C. Four Or More Reference Stations

If a satellite is observed by four or more reference stations ($M \geq 4$), then the synchronized pseudorange residuals are still single differenced to yield

$$\Delta^{(k_1)} = G_W \Delta \mathbf{r}^{(k_1)} + \nu^{(k_1)}.$$

However, this system of equations is either overspecified or exactly specified. In this case, the satellite position estimator is chosen to minimize

$$J_{\text{WMNLS}} = [\Delta \mathbf{r}^{(k_1)}]^T [\Lambda_r^{(k_1)}]^{-1} \Delta \mathbf{r}^{(k_1)} + (\Delta^{(k_1)} - G_W \Delta \mathbf{r}^{(k_1)})^T \cdot P_\Delta^{-1} (\Delta^{(k_1)} - G_W \Delta \mathbf{r}^{(k_1)})$$

where $\Lambda_r^{(k_1)}$ is still the *a priori* covariance of the state, and $P_\Delta = E\{\nu_\Delta^{(k_1)}[\nu_\Delta^{(k_1)}]^T\}$ is the covariance of the measurement noise. The corresponding estimator is given by

$$\begin{aligned} \Delta \mathbf{r}_{\text{WMNLS}}^{(k_1)} &= [(\Lambda_r^{(k_1)})^{-1} + G_W^T P_\Delta^{-1} G_W]^{-1} G_W^T P_\Delta^{-1} \Delta^{(k_1)} \\ &= \Lambda_r^{(k_1)} G_W^T (G_W \Lambda_r^{(k_1)} G_W^T + P_\Delta)^{-1} \Delta^{(k_1)} \\ &= A_W \Delta^{(k_1)}. \end{aligned}$$

The *a priori* information about the satellite location error greatly improves the performance of this estimator. Without

this side information, the least squares solution could yield very noisy estimates of the satellite location error, because the observation geometry is very poor.

Finally, the WMS packs the location, clock, and ionospheric error estimates into the WAAS data format described in the next section.

VII. WAAS MESSAGE FORMAT

For each GPS satellite, the WAAS message contains separate corrections for the quickly varying component of the pseudorange error (mostly satellite clock) and the slowly varying component of the pseudorange error (mostly satellite location). The WAAS message also carries estimates of the ionospheric delay for a grid of locations.

The basic WAAS message is shown in Fig. 19. As shown, all WAAS messages are 250-b in length. At the data rate of 250 bps, the duration of a WAAS message is 1 s, and the start of the message block is synchronous with the 6 s time epoch used by GPS. Each block consists of an 8-b (distributed) preamble, a 6-b message type, a 212-b data field, and a 24-b CRC parity.

The preamble is a 24-b unique word, distributed over three successive blocks. An 8-b preamble is adequate for message synchronization, because the WAAS message is synchronized to GPS time and GPS time is available in the receiver. The message type field is 6-b long, which allows for 64 different messages. The currently defined message types are summarized in Table 1.

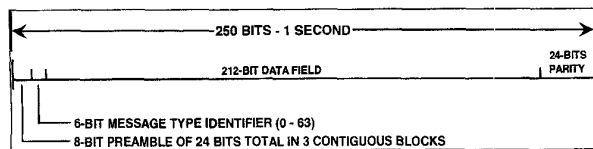


Fig. 19. Basic WAAS data block format.

The WAAS message format serves an extremely ambitious goal: provide Category I precision approach accuracy over a continental area using only a data rate of 250 b/s. Indeed, greater bit rates would require the received signal power from the geostationary WAAS satellites to be greater than the received power from the GPS satellites—an undesirable outcome. Yet the WAAS message stream must carry corrections for all 24 GPS satellites and any GNSS satellites in orbit. In contrast, local area DGPS data links typically use 100 b/s or more to provide a similar capability to ranges of 200 or 300 km. Moreover, local links need only carry corrections for the 6 to 12 satellites which are in view of a single reference station.

The details of the WAAS message format are given in [28], [29]. However, the WAAS message format basically achieves its goal as follows.

1) *Fast and Slow Corrections*: Fast correction messages carry the quickly varying component of the pseudorange errors for each satellite (mostly clock error). They must be sent much more frequently than any other message (every 6–10 s), but they do not decorrelate spatially. Consequently, a single fast correction per satellite suffices for the entire footprint of the WAAS satellite. The GPS error components which vary more slowly are separated and sent much less frequently.

2) *No Rate Corrections*: Unlike most local area DGPS data formats, the WAAS carries no rate corrections for the quickly varying component of the satellite error. As shown in [30], it is more efficient for the user receiver to estimate the rate by differencing the most recent fast corrections.

3) *Masks*: A *mask* is used to designate which satellite belongs to which slot in the fast correction messages. A mask is used to assign slots so that satellite identifications need not be sent with every fast correction. A similar ionospheric mask is used to associate each slot in the ionospheric correction message with a geographic location.

4) *Geostationary Navigation Message*: In contrast to the GPS satellites, the WAAS satellites are geostationary; so their location need only be updated every 2 min or so.

5) *Parity*: The WAAS uses a much stronger parity algorithm than the extended Hamming code used in the GPS navigation message (and typically used for the transmission of differential corrections). The GPS parity algorithm uses the extended Hamming code to add six parity bits to a field of 26 data bits. It achieves a minimum distance of four, which means that any combination of three or fewer bit errors can be detected. However, if an interference or noise burst coincides with one of the 30-b words and the received data is completely garbled, then the probability that the algorithm will fail to detect the error burst is

Table 1 WAAS MESSAGE TYPES

Type	Contents
0	Do not use GEO for anything (for testing)
1	PRN mask assignments, set up to 50 of 212 b
2	Fast pseudorange error estimates
3–8	Reserved for future messages
9	GEO navigation message (x, y, z , time, etc.)
10–11	Reserved for future messages
12	WAAS Network/UTC offset parameters
13–16	Reserved for future messages
17	GEO satellite almanacs
18	Ionospheric pierce point mask 1
19	Ionospheric pierce point mask 2
20	Ionospheric pierce point mask 3
21	Ionospheric pierce point mask 4
22	Ionospheric pierce point mask 5
23	UDRE zone radii and weights
24	Mixed fast/long-term satellite error estimates
25	Long-term satellite error estimate
26	Ionospheric delay error estimate
27–63	Reserved for future messages

$p_{fd} = 2^{-6} = 1.56 \times 10^{-2}$. This probability is too large for the WAAS, because any WAAS message failures must be detected with very high probability—otherwise an integrity failure results.

To reduce the probability of failing to detect a bit error, the parity scheme in Fig. 19 uses 24 b and thus reduces the probability of failing to detect an interference burst to $2^{-24} = 5.96 \times 10^{-8}$. The overhead for parity is reduced from $6/30 = 0.20$ to $24/250 \approx 0.096$. The WAAS error detection algorithm does increase the data latency of the integrity data, since 250 b must be collected by the receiver for a parity check instead of just 30. However, this increases the delay due to transmission time by only 0.88 s at the WAAS data rate of 250 bps.

VIII. USER AVIONICS

All WAAS avionics which are certified by the Federal Aviation Administration (FAA) will conform to the appropriate WAAS Technical Standard Order (TSO). In turn, this TSO will probably be strongly based on a WAAS Minimum Operational Performance Standard (MOPS) currently being developed by the RTCA, which is a federal advisory body comprised of members from industry, government and academia. Specifically, Working Group 2 of RTCA Special Committee 159 is responsible for the WAAS MOPS which is scheduled to be published in early 1996.

Three functional classes of WAAS equipment are defined by the WAAS MOPS. Class β is a WAAS sensor which outputs data to a Flight Management System (FMS). Class γ equipment includes the WAAS sensor and the navigation functions. Finally, Class δ is equipment which is designed to be a drop-in replacement for an ILS receiver. The MOPS also defines classes of WAAS equipment for the various modes of flight from enroute down through precision approach.

As the navigation mode becomes more demanding, the receiver is required to use more and more of the messages in Table 1. Finally, in precision approach, the receiver is

required to apply *all* of the messages shown in Table 1. For this reason, the next subsection describes the pseudorange correction and the navigation solution for precision approach.

A. Pseudorange Corrections and Navigation Solution

During precision approach, the WAAS equipment corrects its measured pseudoranges as follows:

$$\hat{\rho}_u^{(k)} = \rho_u^{(k)} - \Delta \hat{\mathbf{r}}^{(k)} \bullet \mathbf{1}_u^{(k)} + \Delta \hat{B}^{(k)} + (t - t_j) \Delta \hat{B}_u^{(k)} - \hat{I}_u^{(k)} - \hat{T}_u^{(k)} \quad (20)$$

where $\rho_u^{(k)}$ is the measured pseudorange and $\hat{\rho}_u^{(k)}$ is the corrected pseudorange. The correction terms on the right hand side are now discussed.

First, the correction $\Delta \hat{\mathbf{r}}^{(k)} \bullet \mathbf{1}_u^{(k)}$ is the projection of the satellite location error estimate $\Delta \hat{\mathbf{r}}^{(k)}$ onto the line of sight vector from the satellite to the approximate location of the aircraft.

Second, the clock correction is $\Delta \hat{B}^{(k)}$ and as shown the WAAS receiver also develops a rate correction based on recently received corrections as follows:

$$\Delta \hat{B}_u^{(k)} = \frac{\Delta \hat{B}_u^{(k)}(t_j) - \Delta \hat{B}_u^{(k)}(t_{j-1})}{t_j - t_{j-1}}$$

Third, the receiver estimates the vertical ionospheric delay for each satellite by interpolating between the grid point estimates. As shown in Fig. 13, this estimator is linear. Specifically

$$\hat{I}^{(U)} = B_{\text{iono}} \hat{I}^{(G)}$$

The MOPS requires the following weights [21]:

$$b_{j,k} = F(e\ell) \left(\frac{1/d_{k,j}}{\sum_{n=1}^K 1/d_{n,j}} \right)$$

In this case, the subscript k denotes the pierce point for the k th satellite in view of the aircraft. The subscript j is for the j th grid point in the WAAS message. The scale factor $F(e\ell)$ is the obliquity factor given in (17). As shown, the weights also depend on the distance from the k th pierce point to the j th grid point ($d_{k,j}$). These distances are normalized by the sum of the distances in the j th row.

Fourth, the tropospheric correction shown in (20) are developed without the aid of any WAAS provided data, because the tropospheric delays at the reference stations are probably only weakly correlated with the delays at the user. However, a standard tropospheric model is specified in [29], and should remove all but a very small part of the delay.

Finally, the corrected pseudoranges given by (20) are input to the navigation solution which computes the estimates of the user location using the weighted least squares given in (6).

B. Receiver Autonomous Integrity Monitoring

When the WAAS signal is present, all WAAS receivers must be capable of deriving integrity from the WAAS signal. To do this, they must simply check for *do not use* fields contained in the Type 2 and 25 messages. Any satellite flagged as unusable by the WAAS must be immediately deleted from the navigation solution.

All receivers are also required to autonomously monitor integrity to provide safety in the presence of local integrity threats which are not detectable by the WAAS ground segment. These local threats include interference, receiver malfunctions and exceptional multipath events. This autonomous monitoring takes at least four forms.

First, all receivers are required to continuously monitor for steps or discontinuities in their measured pseudoranges. Essentially, this step detector leverages the known dynamics of the aircraft and satellites—it flags as suspect any measurements which are not consistent with the known dynamics.

Second, all receivers are required to prevent interference from causing hazardously misleading navigation information. These algorithms could check to make sure that any tracked signals are modulated with data that corresponds in format to the known structure of the WAAS message or the GPS navigation message. Essentially, they would leverage known properties of these formats to insure that all tracked signals were *bona fide* navigation signals.

The third and fourth autonomous monitors leverage the overspecified nature of the navigation solution. In short, they examine the measurement residuals after the best navigation solution has been computed. If these residuals are large compared to thresholds set based on their normal statistics, then integrity warnings are issued.

More specifically, the third autonomous integrity monitor is used during en route, terminal area, and nonprecision approach as a *reversionary* mode. In the absence of the WAAS signal, the receiver reverts to autonomous fault detection and exclusion (FDE). FDE uses the residuals to not only detect integrity threats, but also to exclude bad measurements from the navigation solution. This algorithm is further described in [29].

Finally, the fourth autonomous integrity monitor is used to fault detect (FD) during precision approach. This algorithm is not reversionary—it must be used during precision approach after the WAAS corrections are applied. In contrast to the reversionary mode described above, this FD algorithm is not required to exclude bad measurements. If it detects an integrity threat, then a missed approach is conducted—no attempt is made to exclude the bad measurement. The remainder of this subsection introduces this *precision fault detection* algorithm, and a more detailed description is given in [31].

As mentioned earlier, the user's navigation solution is almost always overspecified, and the precision fault detection uses this additional information to check on the integrity of the position fix. The vector of measurement residuals is

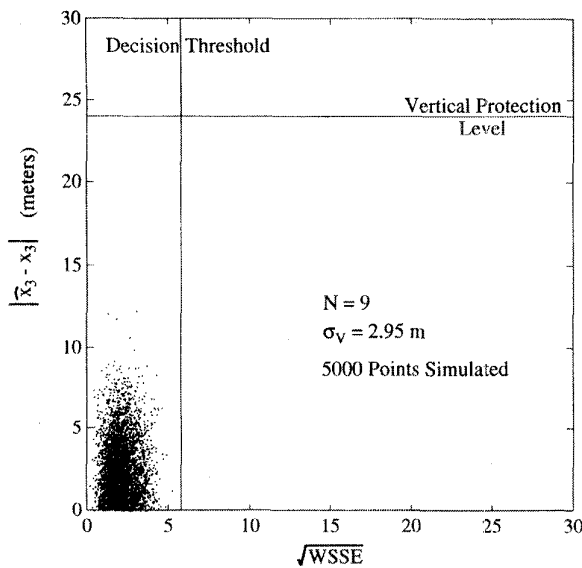


Fig. 20. Vertical positioning error versus the decision statistic with no satellite faults.

given by

$$\begin{aligned}\hat{\epsilon} &= \tilde{\rho}_u - G\hat{x}_u \\ &= (I - GA_U)\tilde{\rho}_u\end{aligned}$$

where $\tilde{\rho}_u$ is the vector of corrected pseudorange measurements and G and A_U are given in (5) and (6), respectively. The user's position and clock solution $\hat{x}_u = A_U\tilde{\rho}$ minimizes a scalar measure given by the weighted sum of the squared errors

$$\begin{aligned}\text{WSSE} &= \hat{\epsilon}^T P_\rho^{-1} \hat{\epsilon} \\ &= [(I - GA_U)\tilde{\rho}]^T P_\rho^{-1} (I - GA_U)\tilde{\rho} \\ &= \tilde{\rho}^T P_\rho^{-1} (I - GA_U)\tilde{\rho}\end{aligned}$$

where P_ρ is the covariance matrix of the corrected pseudoranges. The last equality above is key to what follows and is true essentially because $(I - GA_U)$ is idempotent [32] and P_ρ^{-1} is assumed to have no off-diagonal elements.

$\sqrt{\text{WSSE}}$ is used as the test statistic to judge the goodness of the least squares fit. This statistic is observable, whereas the positioning error $(\hat{x}_u - x_u)$ is not. Fortunately, $\sqrt{\text{WSSE}}$ can be used to reliably flag bad position fixes. A plot of the absolute value of the vertical positioning error $(|\hat{x}_3 - x_3|)$ as a function of $\sqrt{\text{WSSE}}$ is shown in Fig. 20. Normal operation is when the decision statistic $\sqrt{\text{WSSE}}$ and $|\hat{x}_3 - x_3|$ are both small. A false alarm occurs when $\sqrt{\text{WSSE}}$ exceeds the detection threshold, but the vertical position error is acceptable. The $\sqrt{\text{WSSE}}$ threshold is set such that the probability of false alarm is commensurate to the continuity requirement for precision approach. In other words, it is set so that the alarm threshold is exceeded with a very low probability due to noisy measurements when the vertical error is in fact small. The scatter of points in the lower left-hand corner of Fig. 20 are for a simulated set of measurements where no satellite failure has occurred.

A missed detection occurs when $\sqrt{\text{WSSE}}$ is below its threshold, but the vertical error is hazardedly large. This is

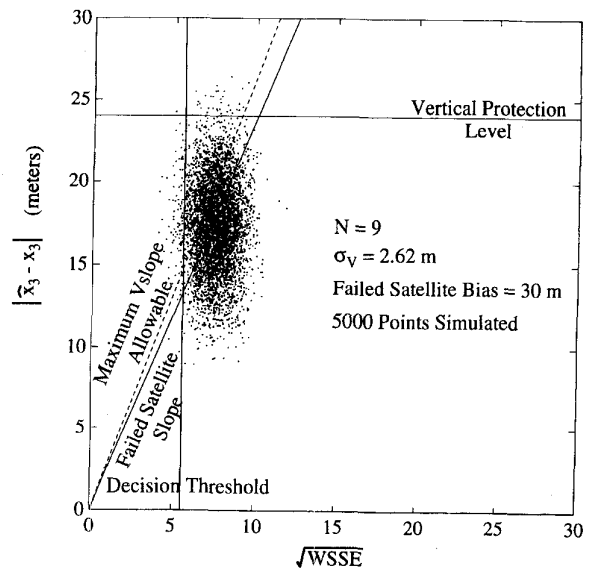


Fig. 21. Vertical positioning error versus the decision statistic with a satellite fault.

the least desirable state since it is an integrity failure—the navigation system has a large error and the pilot is not warned. Finally, a normal detection occurs in the upper right hand quadrant of Fig. 20 when both the decision statistic and the vertical error exceed their respective thresholds.

The ordinary (fault free) distribution of the decision statistic is used to set the alarm threshold (T). When the residuals are normally distributed random variables with zero mean and variance σ^2 , then $\sqrt{\text{WSSE}}$ has a chi-square distribution, and the alarm threshold can be found by numerically inverting the incomplete gamma function [31]

$$1 - P_{FA} = \frac{1}{\Gamma(a)} \int_0^T e^{-s} s^{a-1} ds.$$

In this equation, α is the number of degrees of freedom divided by two, which is equal to

$$\alpha = \frac{K - 4}{2}$$

where as before K is the number of satellite measurements used in the solution. A representative set of thresholds $(T(K, P_{FA}))$ has been computed and tabulated in [31], where it is shown to be a function of the the desired P_{FA} and number of satellites (K) used in the solution.

The probability of missed detection can now be analyzed if we assume that only single satellite failures occur. If this single error introduces a bias b on the k th satellite only, then the measurement error vector is

$$\epsilon = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

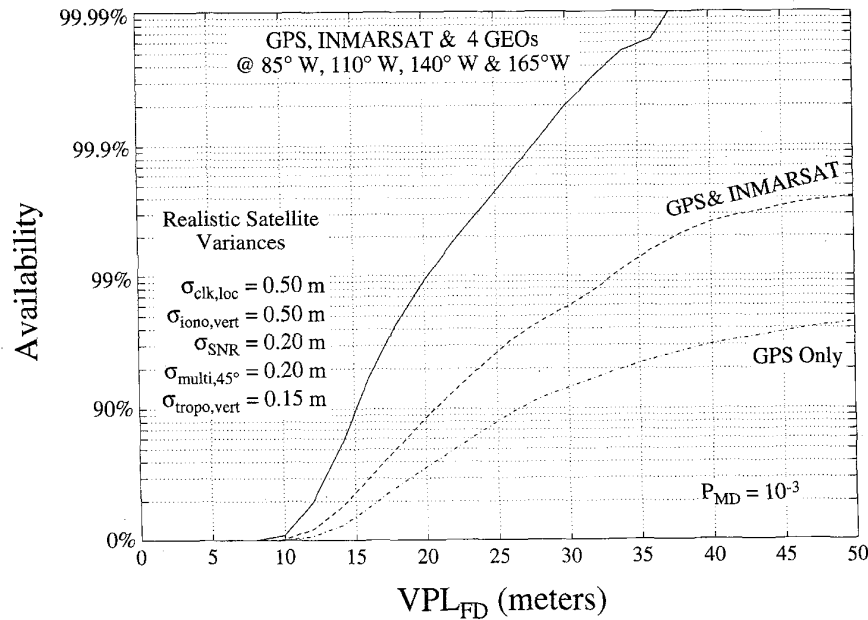


Fig. 22. Time availability of vertical protection level. The vertical protection level VPL_{fd} is the vertical position error which can be detected by the residuals test with a probability of false alarm of 10^{-5} and a probability of missed detection of 10^{-3} .

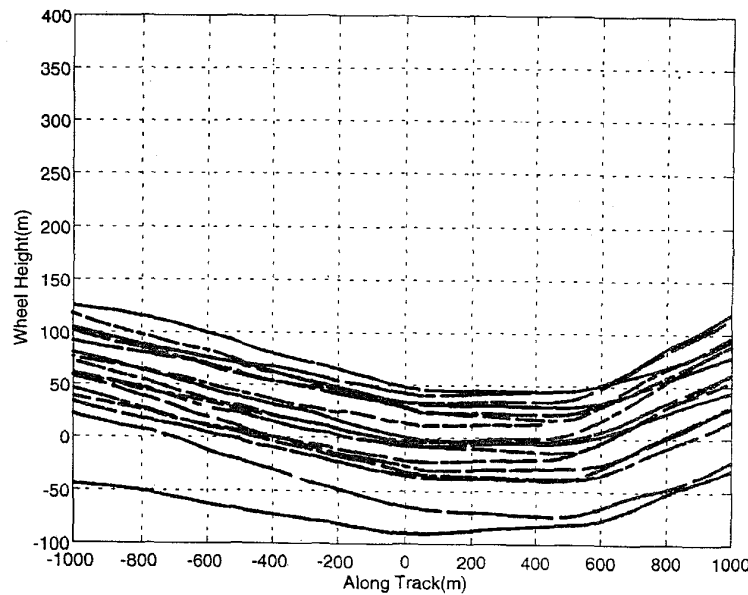


Fig. 23. Vertical position of the aircraft without WAAS corrections. The runway is located at an altitude of 0 m starting at an along track position of 0 m and ending at approximately 700 m.

and the vertical error is

$$\hat{x}_3 - x_3 = A_{U,3,k}b$$

where $A_{U,3,k}$ is the k th element in the third row of A_U . For the same bias error, the decision statistic is

$$WSSE = \frac{b^2(1 - [GA_U]_{k,k})}{(\sigma_p^{(k)})^2}$$

where $[GA_U]_{k,k}$ is the k th element of the k th row of GA_U . The ratio of the magnitude of the vertical error to the test

statistic is

$$\text{slope}_k = \frac{|A_{U,3,k}|\sigma^{(k)}}{\sqrt{1 - [GA_U]_{k,k}}}$$

Note that the value of the bias error b does not effect the value of the slope.

Fig. 21 shows a scatter of simulated measurements when an error bias of $b^{(k)} = 15$ m is included—it contrasts Fig. 20, where no such bias was introduced. As shown, the measurement cluster has moved up a line with slope $^{(k)}$.

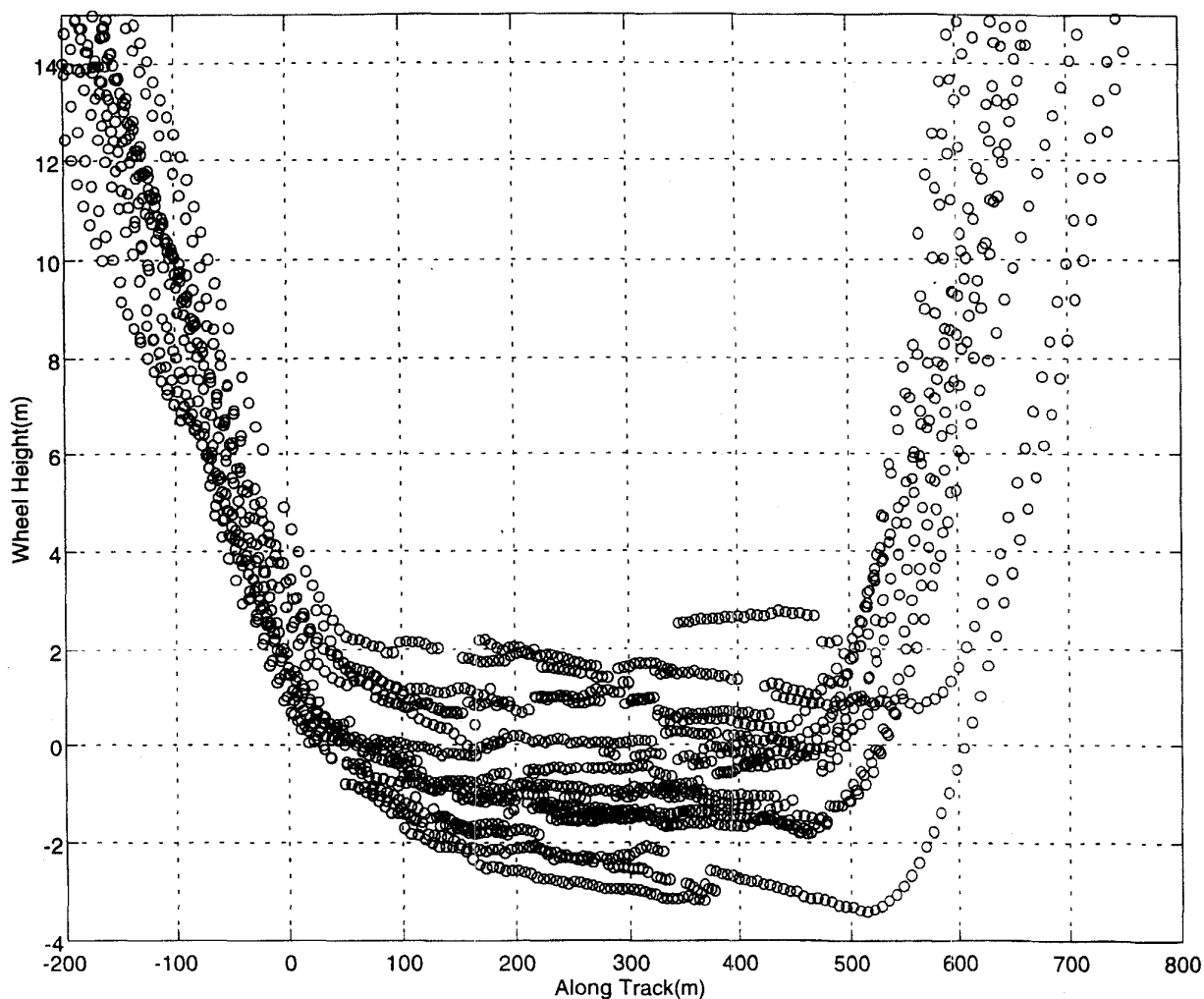


Fig. 24. Vertical position of the aircraft with WAAS corrections. The runway is located at an altitude of 0 m starting at an along track position of 0 m and ending at approximately 700 m.

At this point, the fault detection algorithm can protect the indicated vertical protection level with a given probability of missed detection and probability of false alarm. As shown, the vertical position error which can be detected by the residuals test (VPL_{fd}) depends on the maximum slope, the decision threshold $T(N, P_{fa})$, and the number of standard deviation required to achieve a required probability of missed detection $c_1(P_{md})$. In fact

$$VPL_{fd} = (\text{max slope})T(K, P_{fa}) + c_1(P_{md})\sigma_v$$

where K is the number of satellites used in the solution.

Fig. 22 shows the fraction of time (time availability) that this autonomous capability to protect a specified vertical protection level with a probability of false alarm of 10^{-5} is available. It shows results for the GPS constellation alone, a WAAS constellation of satellites which includes GPS and the four Inmarsat-3 satellites, and finally GPS plus the four Inmarsat-3 satellites plus four additional geostationary satellites across the United States. Both sets of curves include the probability of satellite failures. As shown, the

addition of geostationary satellites improves the availability of autonomous fault detection significantly.

IX. FLIGHT RESULTS

This section gives the results of WAAS flight trials undertaken by Stanford University for the FAA [33]. These use the experimental WAAS shown in Figs. 14 and 15. On average, these experimental WRS's are 660 km from Palo Alto Airport where most of our aircraft trials have occurred.

The reference stations are of the type shown in Fig. 10 and use dial-up phone lines to communicate with the WMS located in the Durand building on the Stanford campus. The WMS algorithms are well-modeled by Figs. 12, 13, and 16.

Our broadcast message is very close in format to the RTCA standard described in Section VII. It sends fast corrections in a Type 2 message, slow corrections in a Type 25 message, and a grid of ionospheric corrections in a Type 26 message.

Our current data link is a cascade of a phone line and a UHF radio, where the radio covers the last few miles to the aircraft. The radio broadcasts 15 W at 462.4 MHz

and operates in broadcast-only mode. It is not configured for repeat transmission requests, and so no acknowledgment are sent from the aircraft to the ground. The radio sends one WAAS message per second. Each message uses 32 bytes or 256 b, where the last 6 b are ignored. The broadcast rate is 9600 bd; so each message requires only 27 ms to send, and the duty cycle is only 2.7%. We are planning to incorporate the ranging function and communications capability of one of the Inmarsat-2 satellites in the near future.

The WAAS data is applied to the raw pseudoranges measured by a single-frequency onboard GPS receiver. In accord with Section VIII, the WAAS data is used to reconstitute scalar pseudorange corrections for each satellite in view of the aircraft. The measured pseudoranges are corrected and used to calculate the aircraft location and glide path deviations.

A set of flight results from the Summer of 1994 are shown in Figs. 23 and 24. The figures show the estimated vertical position of the wheels of a Piper Dakota carrying the WAAS avionics for a set of *touch-and-goes* to Palo Alto airport. Both figures show the estimated vertical position of the plane as a function of its along track position. The along track position of 0 m corresponds to the beginning of the runway at Palo Alto. Fig. 23 is without application of the WAAS corrections, and Fig. 24 includes the corrections. Consequently, Fig. 23 uses a much coarser vertical scale.

Both figures allow an assessment of accuracy because we have very accurately (10 cm) surveyed the location of the runway, and these figures place the runway at 0 m height. Moreover, the aircraft certainly touched down; so the ± 2 m spread shown in Fig. 24 from an along-track location of 100 m–500 m is the error in the WAAS-corrected position. This flight performance is encouraging because it significantly exceeds the 95% accuracy requirement of 7.6 m. Moreover, it is consistent with results from other WAAS flight trials [34], [35].

X. SUMMARY

This paper describes much of the philosophy and design of the WAAS currently being deployed by the FAA. The WAAS augments GPS with the following three services: ranging using "GPS-like" signals from geostationary satellites; accuracy improvements by sending differential corrections using the same signal; and integrity by sending "do not use" warnings over the same signal.

The paper has described a specific set of algorithms for developing the satellite clock and location corrections, as well as the ionospheric corrections. These algorithms are simple, yet a covariance analysis predicts that they will provide the accuracy required for Category I aircraft precision approach. In addition, those predictions have been validated by our early flight tests, which use a small experimental network established by Stanford University for the FAA. Nonetheless, these algorithms are offered as a starting point only—certainly many improvements can be envisioned. For example, models of the forces which act on a satellite in orbit should improve the accuracy and integrity of the satellite location estimates. Additionally,

velocity residuals could be used as the basis for another autonomous integrity check by the aircraft equipment.

In conclusion, the WAAS seems well suited to be the primary navigation system for enroute flight down through Category I precision approach. With certainty, it will find a myriad of other applications as well.

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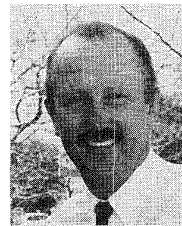
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REFERENCES

- [1] C. Kee, B. W. Parkinson, and P. Axelrad, "Wide area differential GPS," *Navigation, J. U.S. Inst. Navigation*, vol. 38, no. 2, Summer 1991.
- [2] C. Kee, "Wide area differential GPS," Ph.D. dissertation, Stanford Univ., Stanford, CA, Dec. 1993.
- [3] J. Nagle and G. V. Kinal, "Geostationary repeaters: A low cost way to enhance civil user performance of GPS and Glonass," in *Rec. IEEE 1990 Position Location and Navigation Symp.*, Las Vegas, Mar. 1990.
- [4] J. Spilker, "GPS signal structure and theoretical performance," *GPS: Theory and Applications*, B. Parkinson, J. Spilker, P. Axelrad, and P. Enge, Eds. Washington, DC: AIAA, 1996.
- [5] P. Enge, "The global positioning system: Signals, measurements and performance," *Int. J. Wireless Inform. Networks*, vol. 1, no. 2, 1994.
- [6] D. V. Sarwate and M. B. Pursley, "Crosscorrelation properties of pseudorandom and related sequences," *Proc. IEEE*, vol. PROC-68, May 1980.
- [7] B. Forssell, *Radionavigation Systems*. Englewood Cliffs, NJ: Prentice-Hall, 1991.
- [8] A. J. Van Dierendonck and B. D. Elrod, "Ranging signal control and ephemeris/time determination for geostationary navigation payloads," *Proc. 1994 Institute of Navigation Techn. Meet.*, San Diego, CA, Jan. 1994.
- [9] J. M. Durand, T. Michal and J. Bouchard, "GPS availability, Part I," *Navigation*, vol. 37, no. 2, Summer 1990.
- [10] G. V. Kinal, "A note on satellite reliability as applied to the Inmarsat-3 navigation payload," presented to *Working Group 2 of RTCA SC-159*, Jan. 1992.
- [11] RTCA Special Committee 159, Minimum Operational Performance Standards (MOPS) for Airborne Supplemental Navigation Equipment Using GPS, RTCA 204-91/SC159-293 (modified by Technical Standard Order TSO-C129, Dec. 10, 1992).
- [12] P. Enge *et al.*, "Architecture for a civil integrity network using Inmarsat," in *Proc. 3rd Int. Technical Meet. of the Satellite Div. Inst. of Navigation*, Colorado Springs, Sept. 1990.

- [13] C. Cohen *et al.*, "Preliminary results of category III precision landing with 110 automatic landings of a United Boeing 737 using GNSS integrity beacons," in *Proc. Natl. Technical Meet. of the Inst. of Navigation*, Anaheim, Jan. 1995, pp. 157–166.
- [14] R. J. Kelly and J. M. Davis, "Required navigation performance (RNP) for precision approach and landing with GNSS application," *Navigation: J. Inst. Navigation*, vol. 41, no. 1, Spring 1994.
- [15] RTCA Special Committee 159, Minimum Aviation System Performance Standards: DGNSS Instrument Approach System for Special Category I, 1993.
- [16] B. W. Parkinson and P. K. Enge, "Differential GPS," in *GPS: Theory and Applications*, B. Parkinson, J. Spilker, P. Axelrad, and P. Enge, Eds. Washington, DC: AIAA, 1996.
- [17] J. Klobuchar, "Impact of the ionosphere on GPS observables," *GPS World*, 1991.
- [18] C. Goad, "GPS survey," *GPS: Theory and Applications*, B. Parkinson, J. Spilker, P. Axelrad and P. Enge, Eds. Washington, DC: AIAA, 1996.
- [19] J. Spilker, "Tropospheric effects on GPS," *GPS: Theory and Applications*, B. Parkinson, J. Spilker, P. Axelrad, and P. Enge, Eds. Washington, DC: AIAA, 1996.
- [20] M. B. El-Arini *et al.*, "The FAA wide area differential GPS (WADGPS) static ionospheric experiment," in *Proc. 1993 Natl. Techn. Meet. of the Inst. Navigation*, San Francisco, Jan. 1993.
- [21] M. B. El-Arini, J. A. Klobuchar, P. H. Doherty, "Evaluation of the GPS WAAS ionospheric grid algorithm during the peak of the current solar cycle," in *Proc. Institute of Navigation 1994 Natl. Techn. Meet.*, Jan. 1994.
- [22] J. A. Klobuchar, P. H. Doherty, M. B. El-Arini, "Potential ionospheric limitations to wide area differential GPS," in *Proc. 6th Int. Techn. Meet. Satellite Div. of the Inst. Navigation*, Sept. 1993.
- [23] J. A. Klobuchar, "Design and characteristics of the GPS ionospheric time delay algorithm for single frequency users," in *Rec. IEEE 1986 Position Location and Navigation Symp.*, Las Vegas, Nov. 1986, pp. 280–286.
- [24] H. Stark and J. W. Woods, *Probability, Random Processes, and Estimation Theory for Engineers*. Englewood Cliffs, NJ: Prentice-Hall, 1986.
- [25] S. Pullen, P. Enge, and B. W. Parkinson, "A new method for coverage prediction for the Wide Area Augmentation System (WAAS)," in *Proc. Annu. Meet. Inst. Navigation*, Colorado Springs, CO, June 1995.
- [26] J. Ceva *et al.*, "Incorporation of orbital dynamics to improve wide area differential GPS," *Proc. 8th Int. Technical Meet. Satellite Div. Institute of Navigation*, Palm Springs, CA, Sept. 1995, pp. 647–659.
- [27] Y. Tsai *et al.*, "Evaluation of orbit and clock models for real time WAAS," in *Proc. Natl. Techn. Meet. Inst. Navigation*, Anaheim, CA, Jan. 1995, pp. 539–547.
- [28] P. Enge and A. J. Van Dierendonck, "The wide area augmentation system," in *GPS: Theory and Applications*, B. Parkinson, J. Spilker, P. Axelrad and P. Enge, Eds. Washington, DC: AIAA, 1996.
- [29] RTCA Special Committee 159, Minimum Operational Performance Stds. for Airborne Equip. Using Global Positioning Syst./Wide Area Augmentation Syst., D0229, Jan. 1996.
- [30] C. J. Hegarty, "Optimal differential GPS for a data rate constrained broadcast channel," in *Proc. 6th Int. Techn. Meet. Satellite Division of the Inst. Navigation*, Salt Lake City, UT, Sept. 1993, pp. 1527–1535.
- [31] T. Walter, P. Enge, and F. van Graas, "Integrity for the wide area augmentation system," in *Proc. 4th Int. Conf. on Differential Satellite Navigation Syst.*, Bergen, Norway, Apr. 1995.
- [32] R. G. Brown, G. Y. Chin and J. H. Kraemer, "Update on GPS integrity requirements of the RTCA MOPS," in *Proc. 4th Int. Techn. Meet. of the Satellite Division of the Institute of Navigation*, Albuquerque, NM, Sept. 1991, pp. 761–772.
- [33] T. Walter *et al.*, "Flight trials of the Wide Area Augmentation System (WAAS)," in *Proc. 7th Int. Techn. Meet. of the Satellite Division of the Inst. Navigation*, Salt Lake City, UT, Sept. 1994, pp. 1537–1546.
- [34] V. T. Wulfschleger, D. G. O'Laughlin, and F. M. Haas, "FAA flight test results for GPS Wide Area Augmentation System (WAAS) cross-country demonstration," in *Proc. 50th Annu. Meet. Inst. Navigation*, Colorado Springs, CO, June 1994.

- [35] F. M. Haas and M. E. Lage, "GPS Wide Area Augmentation System (WAAS) testbed results—Phase 1D testbed results," in *Proc. 50th Annu. Meet. Inst. Navigation*, Colorado Springs, CO, June 1994.



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